



Universiteit
Leiden

Master Creative Intelligence & Technology
Leiden Institute of Advanced Computer Science
Leiden University

How XAI Explanation Formats Influence Human Judgment in AI-Assisted Misinformation Evaluation

Master's Thesis

Name	Yuning Yao
Student ID	4244435
Date	July 13, 2026
1st supervisor	Derya Soydaner
2nd supervisor	Peter van der Putten

Leiden Institute of Advanced Computer Science
Leiden University
Einsteinweg 55
2333 CC Leiden
The Netherlands

Abstract

AI credibility tools are increasingly used to support judgments about misinformation, but explanation interfaces do not always help users decide when to follow AI advice and when to question it. We examine token-level SHAP attribution cues in an online misinformation-judgment task with five interface conditions: no AI, prediction-only advice, static SHAP, sequential SHAP, and sequential SHAP with a boundary cue. We ask whether these formats change judgment updating, final alignment with AI advice, confidence change, and whether confidence better reflects actual performance.

AI assistance does not produce an overall accuracy benefit. Prediction-only advice and static SHAP show similar final alignment with incorrect AI advice, at about 61%, so static SHAP offers no clear protective advantage over a bare recommendation. Sequential reveal shows fewer descriptive shifts toward incorrect AI advice, while the boundary cue shows lower final alignment with incorrect AI advice but also a possible trade-off when AI advice is correct. Model-based evidence for these condition differences is limited. These findings suggest that evaluating XAI interfaces requires distinguishing between advice-induced updating, final alignment, accuracy, and confidence-related performance, as explanation quality cannot be captured by a single metric.

Contents

1	Introduction	4
1.1	Research objective and questions	5
1.2	Hypotheses	5
1.3	Contributions	6
2	Related Work	6
2.1	Misinformation judgment and truth discernment	6
2.2	Appropriate reliance in human–AI decision making	7
2.3	Behavioral limits of AI explanations	7
2.4	Attribution explanations and interaction design	8
2.5	Research gap and positioning	9
3	Method	9
3.1	Procedure	9
3.2	Study design and problem formulation	10
3.3	Participants	11
3.4	Background questionnaire	11
3.5	Stimuli	11
3.6	AI advisor and explanations	12
3.7	Interface conditions	13
3.8	Counterbalancing	16
3.9	Outcome measures	17
3.10	Statistical analysis plan	18
4	Results	20
4.1	Participant sample and data quality	20
4.2	Overall performance and baseline judgment stability	20
4.3	RQ1: Judgment change across interface conditions	21
4.4	RQ2: Correctness-conditional advice use	22
4.5	RQ3: Confidence and chosen-answer Brier score	25
4.6	Hypothesis-oriented model summary and sensitivity analyses	25
4.7	Item heterogeneity and sensitivity analyses	27
5	Discussion	27

5.1	Principal findings	28
5.2	Prediction-only advice and static SHAP	28
5.3	Sequential reveal, boundary cue, and selectivity	28
5.4	Methodological contribution	29
5.5	Design implications	29
5.6	Limitations	29
5.7	Future work	30
5.8	Implementation, reproducibility, and ethics	30
6	Conclusion	31
	Appendix overview	32
A	Participant background characteristics	32
B	Full background questionnaire items	32
C	Task-flow interface screenshots	33
D	Stimulus items and AI predictions	35
E	Exploratory mixed-effects model details	37
E.1	M1: Final correctness on all trials	38
E.2	M2: Final correctness on AI-present trials	38
E.3	M3: Final alignment on AI-present trials	38
E.4	M4: Final chosen-answer Brier score on AI-present trials	39
E.5	M5: Advice adoption on initial non-alignment trials	39
E.6	Planned contrasts from M3, M4, and M5	39
E.7	Participant-aggregated chosen-answer Brier robustness check	40
E.8	Sensitivity model with displayed model score	40
E.9	Sensitivity check (exclude sessions under 8 minutes)	41
E.10	Exploratory participant-level moderator checks	41

1 Introduction

In online misinformation tasks, AI support often appears as a predicted label with a confidence score, sometimes accompanied by an explanation linked to the prediction. Such support can help users correct an initial mistake, but it can also move users away from a correct judgment when the AI is wrong.

A relevant design objective is therefore not greater agreement with AI. It is appropriate reliance: users should benefit from correct recommendations while resisting incorrect ones (Lee & See, 2004). This distinction is especially important in misinformation evaluation. An intervention may help users reject false content, but it may also make them less willing to accept accurate content (Hoes et al., 2024). Aggregate agreement or accuracy can therefore conceal different effects on correct and incorrect judgments.

Prior work has studied several ways of presenting algorithmic support to users. These include prediction labels and confidence scores as local advice cues (Zhang et al., 2020), model-level accuracy disclosure as expectation setting (de Jong et al., 2025; Yin et al., 2019), feature-attribution explanations such as LIME and SHAP (Lundberg & Lee, 2017; Ribeiro et al., 2016), and interaction techniques such as cognitive forcing or progressive disclosure (Bućinca et al., 2021; Muralidhar, 2024). These approaches share a common design aim: helping users rely on algorithmic advice selectively, accepting useful recommendations while resisting misleading ones.

However, comparable human–AI decision-making work shows that explanations do not necessarily make advice use more sensitive to AI correctness. Explanations can increase acceptance of AI recommendations without making acceptance more selective, and greater transparency can improve users’ ability to simulate model predictions without necessarily helping them detect model errors or make better final decisions (Bansal et al., 2021; Poursabzi-Sangdeh et al., 2021). This means that explanation success should not be evaluated by final agreement alone, but by whether users benefit from correct advice without also following incorrect advice (Schemmer et al., 2023; Schoeffer et al., 2025).

A narrower gap concerns the presentation of explanation content. Much prior work has compared AI advice with and without explanations, or has compared different explanation types. Less attention has been paid to whether the same attribution content produces different judgment-updating patterns when it is presented simultaneously or revealed sequentially. We address this gap by varying the timing and framing of SHAP attribution cues while keeping the item-level AI predictions and underlying SHAP values fixed.

To examine this presentation contrast, participants judge whether short news-style items are true or false before and after exposure to one of five advice-interface conditions: no AI-driven advice, prediction-only advice, prediction plus static SHAP explanation, prediction plus sequentially revealed SHAP explanation, or prediction plus sequential SHAP with a boundary cue. Across the four AI-present conditions, the item-level AI recommendation and SHAP values remain fixed. This lets us compare explanation presentation formats while holding the underlying AI advice and explanation content constant. We treat attribution cues as model-derived interface information, not direct representations of the model’s internal reasoning.

We adopt a decision-dynamics perspective. Final agreement alone does not establish reliance because a participant may already have agreed with the AI before seeing it. Our analysis therefore focuses on changes between initial and final judgments. It distinguishes correction, preservation, and deterioration, and separates responses to correct and incorrect AI advice.

1.1 Research objective and questions

The research objective is to examine how AI explanation presentation shapes selective reliance and confidence–correctness correspondence in AI-assisted misinformation evaluation.

Our main research question is:

How do AI explanation presentation formats affect judgment updating, confidence change, and confidence–correctness correspondence in misinformation evaluation, particularly when AI advice is incorrect?

Our study addresses three sub-questions:

- RQ1.** How do different AI interface formats affect changes from initial to final credibility judgments?
- RQ2.** How do these formats affect movement toward and final alignment with AI advice, especially when the AI recommendation is incorrect?
- RQ3.** How do explanation formats affect confidence change and confidence–correctness correspondence across AI-correct and AI-incorrect trials?

1.2 Hypotheses

Prior work on AI advice, explanations, and interaction design informs the following hypotheses. We use them to organize the analysis and interpretation of the observed judgment-updating patterns.

- H1a: AI advice and judgment change.** We expect AI-present conditions to increase the probability of changing the initial judgment compared with the no-AI baseline.
- H1b: Direction of judgment change.** We expect judgment changes within AI-present conditions to occur more often toward than away from the AI recommendation.
- H2: Static explanations.** We expect static token-level explanations to increase final alignment with the AI recommendation compared with prediction-only advice.
- H3: Incorrect-AI alignment.** We expect static explanations to increase final alignment with incorrect AI recommendations compared with prediction-only advice.
- H4: Sequential reveal.** We expect sequential reveal to reduce movement toward incorrect AI advice and final alignment with incorrect recommendations compared with static explanations.
- H5: Boundary cue.** We expect the boundary cue to reduce final alignment with incorrect AI recommendations and limit confidence increases on incorrect final judgments compared with sequential reveal alone. We do not expect it to substantially reduce alignment when the AI is correct.

We report descriptive trial-level patterns and use mixed-effects models as the principal inferential analyses. We interpret the model results cautiously because the study has a modest number of AI-incorrect trials per condition and does not directly measure processing fluency, cognitive effort, perceived understanding, attention allocation, or awareness of model limitations. We therefore treat mechanism claims as interpretations rather than tested mediators.

1.3 Contributions

The thesis makes four contributions.

1. We compare static and sequential presentations of the same token-level SHAP attribution content in an AI-assisted misinformation judgment task. This allows us to examine whether changing presentation format, while holding prediction and attribution values constant, relates to different judgment-updating patterns.
2. We combine initial-to-final judgment transitions with correctness-conditional reliance measures. This avoids treating final agreement with AI as reliance when participants may already have held the AI-aligned judgment before advice exposure.
3. We provide descriptive evidence that interface conditions can show different outcome profiles across harmful shifts, final incorrect-AI alignment, beneficial reliance, and confidence–correctness correspondence. Researchers therefore need to evaluate explanation formats with more than one global agreement or accuracy metric.
4. We make the study code and analysis data available through a public GitHub repository.¹

2 Related Work

This section reviews the work needed to motivate the study design. It first discusses misinformation judgment as a truth-discernment problem, then reviews appropriate reliance in human–AI decision making, behavioral limits of AI explanations, and interaction-design choices for presenting attribution cues.

2.1 Misinformation judgment and truth discernment

Misinformation evaluation requires users to distinguish true from false content rather than becoming uniformly more skeptical. This distinction is commonly described as truth discernment. Changes in overall belief do not necessarily indicate improved discernment because an intervention may reduce belief in both true and false information (Pennycook & Rand, 2021).

Misinformation research identifies several routes through which discernment can fail. Cognitive, social, and affective factors shape belief in misinformation, including familiarity, prior beliefs, source cues, and emotional responses (Ecker et al., 2022). Lower analytic reasoning is associated with greater susceptibility to partisan false news, and poor discernment is also linked to limited relevant knowledge and familiarity-based heuristics (Pennycook & Rand, 2019, 2021).

The same logic applies to intervention design. Accuracy incentives improve truth discernment, whereas incentives tied to political in-group approval reduce it (Rathje et al., 2023). Inoculation against emotional manipulation improves manipulation detection, but improves truth discernment only when paired with an accuracy prompt (Pennycook et al., 2024). Several misinformation interventions can also reduce belief in false information while lowering the perceived credibility of factual information (Hoes et al., 2024). These findings matter because an intervention can appear successful if only false-item belief is measured, while still weakening acceptance of true information.

¹<https://github.com/itsyuning/xai-ui-study>

Correction and warning cues raise a related measurement problem. Political corrections can fail among the subgroup most likely to hold a misperception and can sometimes backfire, although later work cautions against treating strong backfire effects as the usual outcome (Nyhan & Reifler, 2010; Swire-Thompson et al., 2020). Recent misinformation-interface work similarly treats warning labels as behavioral interventions whose effects depend on presentation and context (Gruzd et al., 2024). For our analysis, the requirement is clear: we evaluate AI support separately on true and false items, so that correction, preservation, and deterioration remain visible rather than being absorbed into an overall belief score.

2.2 Appropriate reliance in human–AI decision making

The goal of AI-assisted decision making is not maximal trust or agreement, but appropriate reliance. Lee and See (2004) argue that users should rely on automation in ways that reflect its capabilities and limitations. Trust may influence reliance, but the two constructs are not equivalent. Hoff and Bashir (2015) further show that trust depends on dispositional, situational, and learned factors.

Inappropriate automation use can take opposite forms: misuse and disuse (Parasuraman & Riley, 1997). Evidence from automated-aid and algorithm-advice settings shows why this balance is unstable. Observed aid errors can reduce trust and lead users to ignore even superior automated aids (Dzindolet et al., 2002). Observed algorithm errors can also reduce willingness to use algorithms, even when the algorithm remains superior to a human forecaster (Dietvorst et al., 2015). In other numerical-estimation tasks, participants weight algorithmic advice more strongly than comparable human advice (Logg et al., 2019). These findings caution against treating reliance as a stable user tendency; it shifts with task context and prior performance information.

Case-specific studies provide a stronger basis for evaluation. Zhang et al. (2020) find that model confidence helps users differentiate cases with different levels of model certainty, but improved trust calibration does not automatically improve joint performance. Related work on accuracy disclosure shows that communicating model-level accuracy can also shape reliance, but this differs from showing an item-level prediction score because it sets a general expectation about the system rather than explaining a specific recommendation (de Jong et al., 2025).

Schoeffer et al. (2024) define reliance through adherence to or override of AI recommendations, conditioned on recommendation correctness. Their results show that explanations can change override behavior without improving users’ ability to distinguish correct from incorrect advice.

Chaleshtori et al. (2024) emphasize the need to record an initial unaided judgment. Final agreement with AI does not necessarily indicate reliance because users may already have held the same judgment. We use the initial response as part of the outcome structure, which lets us distinguish correction, preservation, and deterioration.

2.3 Behavioral limits of AI explanations

Human-centered XAI research argues that evaluations should judge explanation quality relative to a specific task and user rather than infer it from technical properties alone (Abdul et al., 2018; Doshi-Velez & Kim, 2017).

Behavioral evaluations complicate the assumption that explanation presence improves decision quality. Explanations can increase acceptance of AI recommendations regardless of correctness, without improving performance beyond showing the recommendation and confidence (Bansal et al., 2021). Greater transparency can also improve model simulatability without consistently improving error detection or

model use (Poursabzi-Sangdeh et al., 2021).

Across tasks, explanation effects appear conditional rather than uniformly beneficial. Benefits emerge only in limited method–task combinations, and subjective explanation ratings do not reliably predict simulatability (Hase & Bansal, 2020). Explanation effects also vary with task knowledge, model complexity, explanation type, representation, and conceptualization, which can affect comprehension, confidence, trust, and reliance in different ways (Delaunay et al., 2025; Pareek et al., 2024; Wang & Yin, 2022).

User knowledge shapes how people interpret explanations. Chen et al. (2023) identify three forms of intuition used to evaluate AI support: intuition about the task outcome, relevant features, and AI limitations. In their experiments, example-based explanations produce less overreliance than the feature-based explanations tested, although the result does not generalize beyond those tasks by itself.

Technical experience also does not guarantee correct interpretation. Kaur et al. (2020) find that data scientists using InterpretML and SHAP sometimes over-trust and misuse the tools. Morrison et al. (2025) further show that explanation correctness can vary independently of prediction correctness and can affect reliance and team performance.

The common thread is that explanations often change acceptance or perceived understanding without reliably improving judgment quality. We target this behavioral gap by evaluating explanation formats through judgment transitions, error detection, and final performance rather than through explanation presence or subjective clarity alone.

2.4 Attribution explanations and interaction design

We use token-level SHAP attributions because the experiment requires stable, rankable textual cues. We can present these cues in different formats while holding explanation content constant. SHAP provides additive feature attributions grounded in Shapley values (Lundberg & Lee, 2017). This makes it suitable for constructing both a static heatmap and a sequential reveal from the same attribution values.

Researchers should not treat SHAP outputs as direct representations of model reasoning. Jacovi and Goldberg (2020) distinguish plausibility to users from faithfulness to model behavior. Adebayo et al. (2018) show that some saliency methods fail sanity checks, while Slack et al. (2020) demonstrate that post-hoc explanations can be manipulated to conceal undesirable model behavior.

Automatic explanation metrics also do not guarantee human usefulness. Nguyen (2018) reports only moderate correspondence between deletion-based metrics and human judgments of local text explanations. Chaleshtori et al. (2024) similarly argue that realistic decision tasks with strong baselines should test explanation utility. In their legal claim-verification study, highlights and influential examples do not consistently improve decisions beyond AI predictions and confidence.

Interaction design may change how participants use explanation cues. Buçinca et al. (2021) find that cognitive-forcing interfaces reduce overreliance, but also receive less favorable subjective evaluations. Their results show that reducing overreliance may require additional effort.

One relevant design choice is whether users form an initial judgment before seeing AI advice. This sequencing preserves an observable unaided response. It also lets us distinguish later correction, preservation, and deterioration. Sequentially revealing explanation cues is not equivalent to prior cognitive-forcing interfaces. Delayed advice, user-controlled disclosure, waiting requirements, and incremental feature reveal impose different forms of effort.

Prior work explores progressive disclosure as a transparency-oriented interface principle (Muralidhar, 2024). The available evidence relies mainly on prototype evaluation and subjective feedback. It provides

a design rationale for staged presentation, but not established evidence that sequential explanations improve reliance, confidence–correctness correspondence, or accuracy.

Sequential reveal may produce competing effects. Presenting attribution cues incrementally may reduce simultaneous information load and encourage smaller evidence evaluations. Conversely, it may increase anchoring on the earliest or strongest cues. This unresolved tension motivates the C2–C3 comparison: the two conditions use identical attribution information but organize it differently over time.

2.5 Research gap and positioning

The reviewed work leaves a specific gap. Explanations can change reliance without improving sensitivity to AI correctness, and their effects depend on user knowledge, explanation type, model properties, and interface design.

Most studies compare different explanation methods or explanation presence versus absence. The temporal presentation of identical attribution information has received less direct attention. The contrast is important here because a static heatmap and a sequential reveal can contain the same attribution values while creating different interaction patterns.

We isolate this presentation contrast by holding item-level AI predictions and SHAP attribution values constant across C2 and C3. We also examine whether adding a boundary cue to sequential reveal relates incrementally to different outcomes in C4. The analysis treats C4 as sequential reveal plus an added cue, not as an independent explanation format.

We do more than compare explanation with no explanation. We record participants’ initial judgments, the direction of any judgment change, and whether the AI recommendation is correct. We evaluate judgment transitions, correctness-conditional alignment, deterioration, final accuracy, confidence change, and confidence–correctness correspondence. We do not treat increased agreement, greater perceived transparency, or higher confidence as sufficient evidence of explanation success. Instead, we ask whether updating remains sensitive to whether the AI recommendation is correct.

3 Method

We use a within-subject experimental design to examine how different AI explanation formats affect human credibility judgments. Stimulus groups are counterbalanced across the five interface conditions to limit systematic order effects and item-condition confounds. We combine a staged judgment task, fixed item-level AI advice, and trial-level measures of judgment updating and confidence.

3.1 Procedure

Participants complete the study online in two stages. They first open a Qualtrics survey, which begins with an information and consent page. Participants who consent complete the background questionnaire. At the end of the questionnaire, Qualtrics redirects them to the custom web experiment.² The redirect applies the counterbalancing-list assignment described later in this section.

In the web experiment, each participant completes 25 trials. On each trial, participants read a short news-style item and select whether they believe it is TRUE or FAKE. They then rate confidence in that initial judgment. They next enter the interface stage. In C0, the interface shows no AI information, but

²Public demonstration deployment: <https://xaiui.vercel.app>. This link documents the interface flow. The analyses use the archived matched participant-task dataset described in the Method and Results sections.

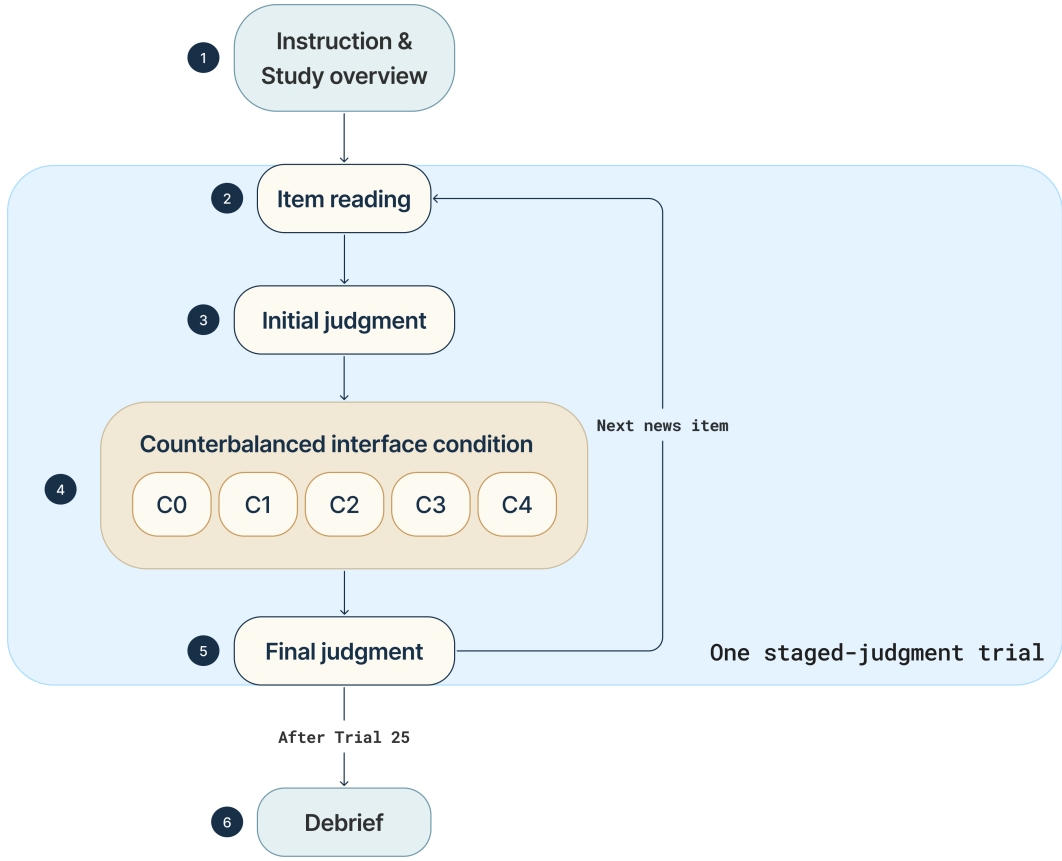


Figure 1: Overview of the web-experiment flow after the Qualtrics survey redirect. Numbered markers indicate (1) instruction and study overview, (2) item reading, (3) initial label and confidence judgment, (4) counterbalanced interface condition, (5) final label and confidence judgment, and (6) debrief. The loop indicates repetition across 25 news items before the debrief page.

participants still continue to the final judgment stage. In AI-present conditions, participants see the AI prediction and, depending on the condition, explanation cues or a boundary cue. After the interface stage, all participants make a final TRUE/FAKE judgment and again rate their confidence. The task presents the 25 trials in randomized order within each participant’s assigned counterbalancing list, so condition trials are intermingled rather than blocked.

After the final trial, participants see a debrief page. The debrief explains why the study includes misleading content and imperfect AI advice, provides contact information, and offers optional item clarification.

Figure 1 summarizes the web-experiment flow after the Qualtrics survey redirect. Appendix C provides screenshots of the instruction, item reading, initial judgment, and final judgment stages.

3.2 Study design and problem formulation

We operationalize AI-assisted misinformation evaluation as judgment updating under external advice. Each trial consists of three stages. First, participants read a short news-style item and make an initial

true/false judgment. They also report confidence in that judgment. Second, depending on the assigned condition, they either reconsider the item without AI assistance or see an AI prediction with one of several explanation formats. Third, participants make a final true/false judgment and again report confidence.

Formally, each trial represents a transition from an initial human judgment $J_{ij}^{(0)}$ to a final human judgment $J_{ij}^{(1)}$ after exposure to an interface condition C_{ij} , where i indexes participants and j indexes trials. In AI-present conditions, the participant also receives an AI recommendation A_{ij} . This structure lets us measure whether the participant stays with their initial judgment, shifts toward the AI, shifts away from the AI, or aligns with the AI at the final judgment stage.

The staged design is necessary because we do not treat AI assistance as a simple performance intervention. Instead, we treat AI output as an advisory signal that may relate to changes in the participant’s judgment trajectory. With this design, we ask not only whether $J_{ij}^{(1)}$ is correct, but also how $J_{ij}^{(1)}$ differs from $J_{ij}^{(0)}$, whether this difference follows the AI recommendation, and how confidence changes after the interface stage. The same initial-to-final structure in C0 helps separate AI-related updating from ordinary reconsideration.

3.3 Participants

Participants were recruited online through convenience sampling via personal networks, student communities, and relevant online forums. Eligible participants were adults who could read English and complete the study remotely. Participants first accessed a Qualtrics survey containing the study information, consent form, and background questionnaire. After completing the questionnaire, they were redirected to the custom web task.

The task was designed to include 25 trials per participant, with five trials assigned to each interface condition through the counterbalanced rotation described above. Participant background variables were collected to describe the sample and support interpretation of the results, not as primary predictors in the main hypothesis tests.

3.4 Background questionnaire

The background questionnaire collects demographic and task-relevant variables. These include age group, gender, country or cultural background, first or primary language, education level, English reading comfort, and online news-reading frequency. They also include familiarity with misinformation and fact-checking, perceived usefulness of AI-generated predictions or explanations, likelihood of verifying AI advice, and general caution toward AI-generated outputs. We export questionnaire responses from Qualtrics and link them to task records using anonymized participant IDs. The full questionnaire items are provided in Appendix B.

We use the questionnaire to describe the final participant sample and check whether participants have sufficient English reading comfort for the task. These variables are not part of the experimental manipulation, so we treat them as descriptive background variables.

3.5 Stimuli

The stimulus set consists of 25 short news-style items from a curated item pool that we prepare before data collection. We collect candidate items from public fact-checking and news-analysis sources,

including AFP Fact Check, Snopes, PolitiFact, and other topic-specific institutional outlets. We then select items to cover politics, health, science and technology, business, and society. Each item includes a headline-like title and a short content summary. Each item has a fixed ground-truth label, either TRUE or FAKE, based on the originating source material or dataset label. Appendix D provides the full list of item titles, topic categories, ground-truth labels, AI predictions, and item-level source provenance.

We keep the stimuli short enough for repeated trial-based evaluation while still supporting a credibility judgment. This choice reflects online misinformation contexts, where users often encounter condensed claims, headlines, summaries, or short posts rather than full articles. The format also imposes a limitation. Some claims may require background knowledge, source checking, or visual evidence that is not fully available in the short text. The Discussion returns to this limitation.

The final stimulus pool includes both AI-correct and AI-incorrect cases because appropriate reliance means following AI advice when it is correct and resisting it when it is wrong. We purposively construct the pool to contain a relatively high proportion of AI errors (10 of 25 items). This enables analysis of behavior under incorrect advice. We therefore do not interpret aggregate accuracy and reliance rates as estimates of performance under the model’s natural error distribution.

3.6 AI advisor and explanations

The AI advisor is a misinformation-classification system that produces a binary recommendation for each item. We use a publicly available Hugging Face checkpoint based on the RoBERTa architecture (Liu et al., 2019): `hamzab/roberta-fake-news-classification`.³ Participants see the AI output as either TRUE or FAKE. The AI advisor uses a fixed binary classification threshold. It classifies an item as FAKE when $P(\text{FAKE}) \geq 0.5$, and as TRUE otherwise:

$$\hat{y}_{AI} = \begin{cases} \text{FAKE}, & \text{if } P(\text{FAKE}) \geq 0.5, \\ \text{TRUE}, & \text{otherwise.} \end{cases}$$

We used a fixed 0.5 threshold, equivalent to the standard argmax rule for a binary softmax classifier: items were labelled FAKE when $P(\text{FAKE}) \geq P(\text{TRUE})$. The checkpoint was fine-tuned on the Kaggle Fake and Real News dataset,⁴ whose classes are approximately balanced; this does not imply equal real-world prevalence of misinformation and accurate news. The same threshold was applied to all stimuli before participant data collection. In AI-present conditions, the interface displays this recommendation after the participant makes an initial judgment. We use the same item-level AI label and displayed model score across AI-present conditions. The displayed model score is the model probability for the displayed binary label: $P(\text{FAKE})$ for FAKE recommendations and $1 - P(\text{FAKE})$ for TRUE recommendations. It represents model output, not an independently calibrated probability. The experimental manipulation concerns the presence and presentation of explanation cues and the boundary cue. Appendix D reports the fixed item-level AI predictions. The study interface was implemented as a static web application using HTML, CSS, and JavaScript, and deployed through Vercel. Python scripts and a notebook using PyTorch, Hugging Face Transformers, and SHAP generated the fixed model predictions and token-level attribution values before data collection.

For explanation-based conditions, the interface uses token-level attribution cues that we compute with SHAP, an additive feature-attribution method based on Shapley values (Lundberg & Lee, 2017).⁵ The interface presents these cues as highlighted textual features associated with the AI prediction. The goal

³Model card: <https://huggingface.co/hamzab/roberta-fake-news-classification>

⁴<https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset>

⁵SHAP software repository: <https://github.com/shap/shap>

is not to evaluate the technical faithfulness of SHAP itself, but to examine how different presentation formats of otherwise fixed attribution cues affect human judgment updating.

This design separates the model’s prediction from the interface format for presenting the prediction and explanation. The same general type of AI advice can become more or less influential depending on whether the interface shows it alone, pairs it with static highlights, reveals it sequentially, or accompanies it with a boundary cue.

3.7 Interface conditions

The independent variable is the AI interface condition. We use five conditions. Figure 2 shows the full-screen layout using a C3 sequential-reveal trial as an example. Figures 3, 4, 5, 6, and 7 then show the condition-specific interface stages.

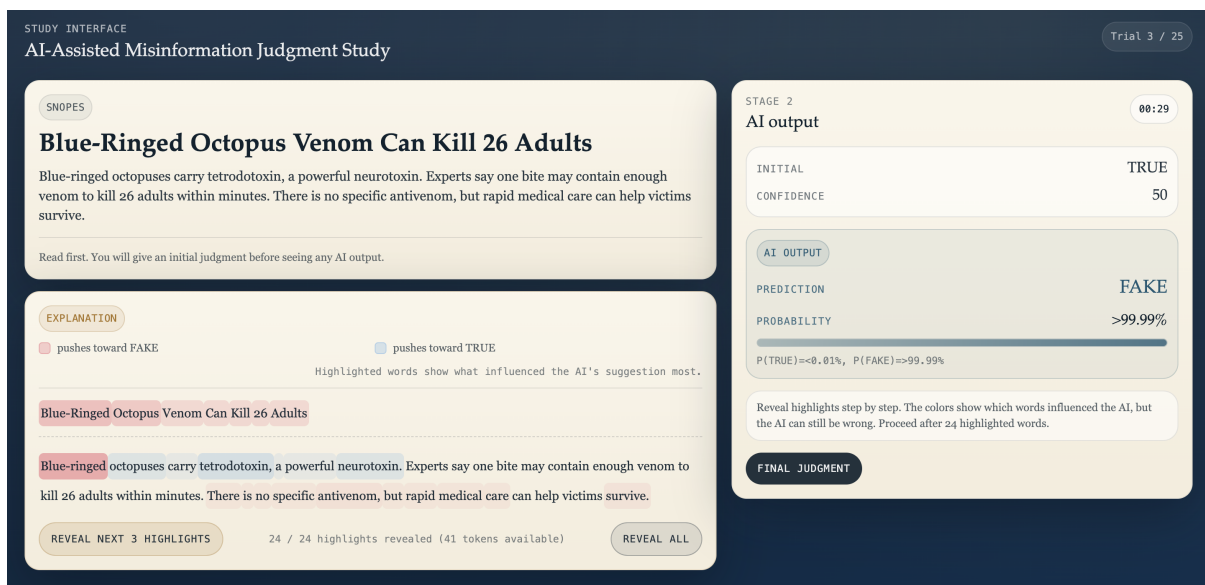


Figure 2: Full-screen example of the study interface in the C3 sequential-reveal condition. The example shows the stimulus text, the participant’s initial judgment, the AI output, and the token-level explanation panel after all highlights have been revealed.

C0: No AI baseline. Participants make an initial judgment and a final judgment without receiving AI advice. We include this condition to measure natural judgment stability, reconsideration, and confidence change when no new AI information is introduced. C0 is therefore a repeated-judgment baseline, not a reliance condition.

STAGE 2

00:13

Review (no AI)

INITIAL

TRUE

CONFIDENCE

67

Review screen: re-read the headline/excerpt. No AI is available for this trial. Proceed when ready.

FINAL JUDGMENT

Figure 3: C0 no-AI baseline interface. Participants complete the same staged judgment task without seeing an AI recommendation.

C1: Prediction-only. Participants see the AI's binary prediction, either TRUE or FAKE, together with the displayed model score. They do not see token-level highlights or a boundary cue. This condition isolates the influence of the AI recommendation itself and provides the comparison point for testing whether explanation cues add anything beyond prediction-only advice.

STAGE 2

00:25

AI output

INITIAL

TRUE

CONFIDENCE

50

AI OUTPUT

PREDICTION

TRUE

PROBABILITY

>99.99%

P(TRUE)=>99.99%, P(FAKE)=<0.01%

AI label + probability are shown. Proceed when ready.

FINAL JUDGMENT

Figure 4: C1 prediction-only interface. The interface shows the AI recommendation and displayed model score without token-level explanation cues.

C2: Static explanation. Participants see the AI prediction and displayed model score together with a static token-level explanation. The explanation appears as highlighted textual evidence associated with the AI output.

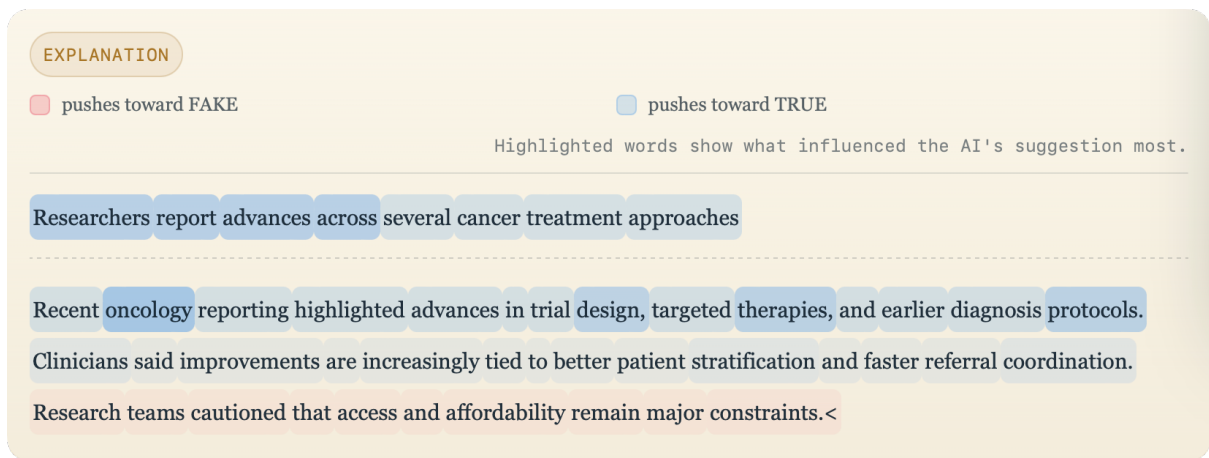


Figure 5: C2 static explanation interface. All token-level SHAP highlights appear simultaneously with the AI recommendation.

C3: Sequential reveal. Participants first see the AI prediction without the full explanation. They then click the *Reveal Next 3 Highlights* button to reveal three additional token-level SHAP highlights at each step until all 24 explanation cues become visible. Highlights are revealed in descending order of absolute SHAP value, so participants first see the tokens with the strongest influence on the model output. This condition tests whether incremental presentation affects judgment updating relative to displaying all highlights simultaneously.

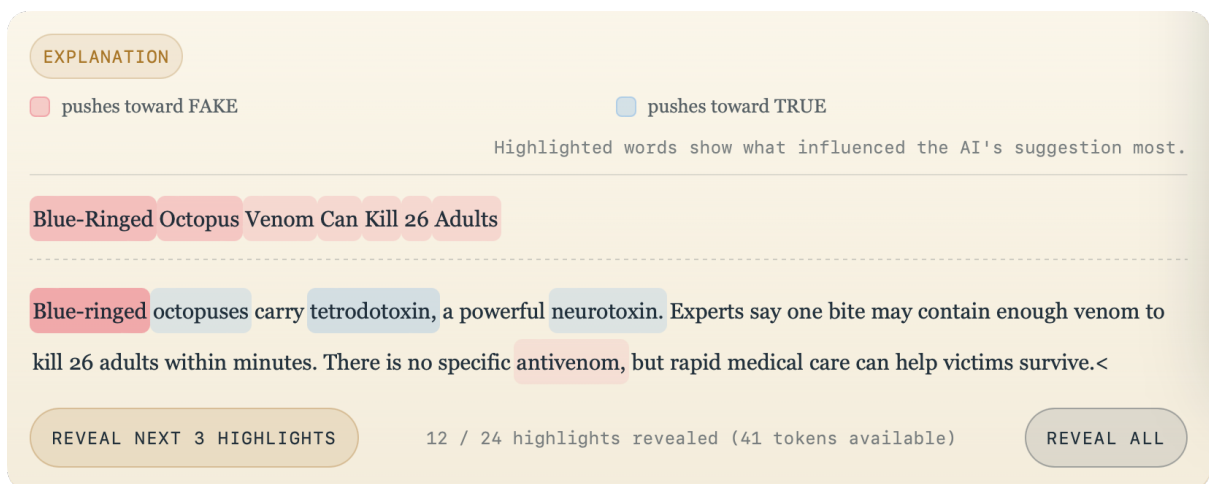


Figure 6: C3 sequential-reveal interface. Participants reveal token-level SHAP highlights in steps rather than seeing all highlights at once.

C4: Sequential reveal with boundary cue. Participants see the sequential explanation interface together with an explicit cue reminding them that the AI can be wrong. This condition tests whether making model fallibility salient relates to lower incorrect-AI alignment or better confidence–correctness correspondence under AI error.

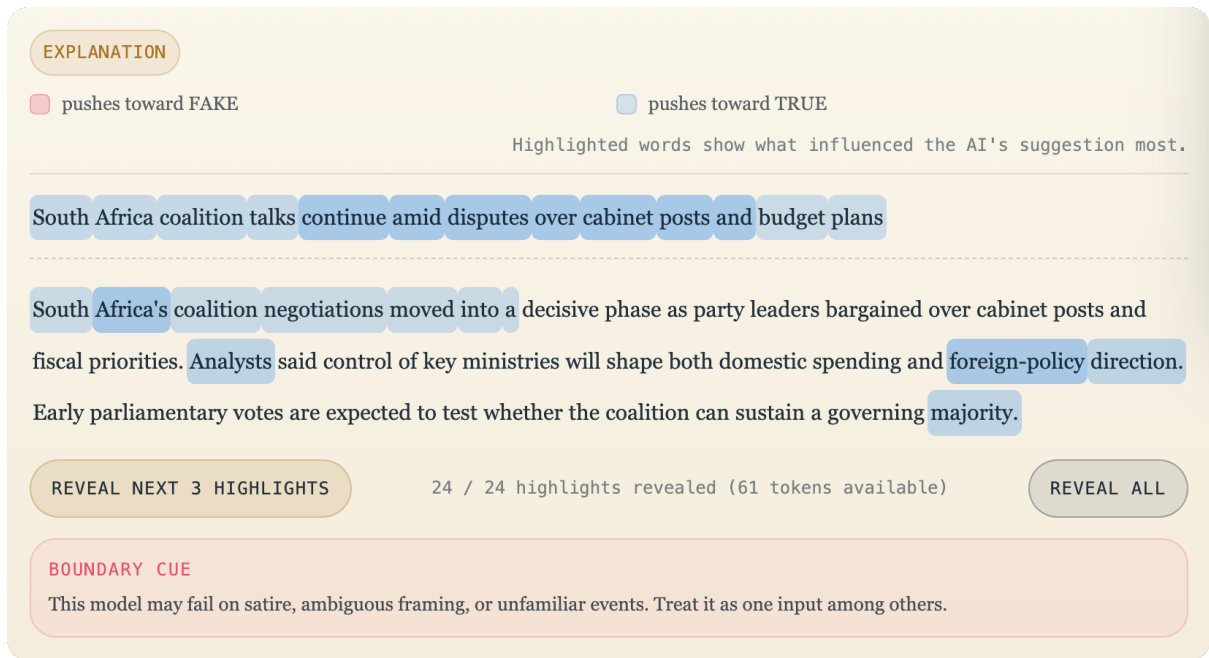


Figure 7: C4 sequential-reveal interface with boundary cue. The cue reminds participants that the AI may fail on satire, ambiguous framing, or unfamiliar events.

These five conditions distinguish prediction exposure, explanation exposure, interaction pacing, and explicit uncertainty framing. C0 asks how much judgment and confidence change when participants reconsider the same item without AI input. C1 asks how much changes once the AI prediction and displayed model score are introduced. The comparison between C1 and C2 tests whether static explanation changes advice use beyond prediction-only advice. The comparison between C2 and C3 tests whether sequential presentation changes advice-use dynamics. The comparison between C3 and C4 tests whether an explicit boundary cue relates to lower incorrect-AI alignment beyond sequential reveal. In C0, AI correctness is an item-level counterfactual property used only for matched accuracy comparisons. It does not represent advice exposure or reliance.

3.8 Counterbalancing

We used a counterbalanced within-subjects design to meet three requirements. First, each participant needed to experience all five interface conditions so that condition differences could be examined within participants. Second, each news item could appear only once for a participant, because repeated exposure to the same item would make the second judgment less independent. Third, item properties such as topic, difficulty, plausibility, and AI correctness should not be tied to a single interface condition. Before data collection, we therefore divided the 25 news items randomly into five fixed item groups (A–E), with five pre-specified items in each group. These item groups did not change across participants. We then created five counterbalancing lists. A list refers to one full 25-trial version of the task, in which each item group is assigned to one of the five interface conditions. Each participant was assigned to one list only, so each participant completed five trials per condition and 25 trials in total.

The group-to-condition assignment was rotated across five lists. Thus, each participant saw each news item only once, but across the five lists each item group appeared once in every interface condition. This allowed us to compare interface conditions within participants while reducing the risk that item-specific properties, such as topic, difficulty, or plausibility, were confounded with a particular condition.

Table 1: Item-group-to-condition mapping in the five-list counterbalancing scheme. Each participant completed only one list. Within a list, each item group is assigned to a different interface condition, resulting in five trials per condition and 25 trials per participant. Across lists, each item group appeared once in every condition.

List	Group A	Group B	Group C	Group D	Group E	<i>n</i>
List 1	C0	C1	C2	C3	C4	14
List 2	C1	C2	C3	C4	C0	14
List 3	C2	C3	C4	C0	C1	14
List 4	C3	C4	C0	C1	C2	14
List 5	C4	C0	C1	C2	C3	14
Total						70

Note. C0 = no-AI baseline; C1 = prediction only; C2 = static explanation; C3 = sequential reveal; C4 = sequential reveal with boundary cue.

Table 1 shows the item-group-to-condition mapping. The final matched sample contained 14 participants in each list, for a total of 70 participants.

AI correctness was also distributed across item groups, but not perfectly evenly. The groups contained different numbers of AI-wrong items: A = 2, B = 2, C = 3, D = 2, and E = 1. Because groups rotated across conditions, the number of AI-wrong trials in a given condition varied by list. Each participant still completed five trials per condition, but a condition could contain between one and three AI-wrong trials depending on the assigned list. Aggregated across the evenly filled lists, each AI-present condition contained 140 AI-wrong trials and 210 AI-correct trials.

The implemented web task randomized trial order within each participant’s assigned list using a seed-based order. This intermingled trials from different conditions rather than presenting conditions in blocks. The design therefore counterbalanced item-to-condition assignment, but did not fully counterbalance condition order; condition order was not treated as a primary target of the study.

3.9 Outcome measures

We use several trial-level outcome measures to capture decision dynamics.

Initial and final accuracy. Initial accuracy indicates whether the participant’s first judgment matched the ground-truth label. Final accuracy indicates whether the participant’s final judgment matched the ground truth after the interface stage.

Judgment shift. A judgment shift occurs when the participant’s final judgment differs from their initial judgment. This measure captures whether the participant changes their mind during the trial.

Final AI alignment. In AI-present conditions, final AI alignment indicates whether the participant’s final judgment matched the AI recommendation. We do not treat alignment as inherently good or bad. It helps when the AI is correct and becomes problematic when the AI is incorrect. We interpret final alignment on AI-incorrect trials as final alignment with incorrect AI advice, not as evidence that AI exposure caused that alignment. Some participants already agreed with the AI before exposure. Trials in which an initially correct response changed to an incorrect response matching the AI recommendation provide stronger evidence of harmful advice adoption.

Override. In AI-present conditions, override indicates whether the participant’s final judgment does not match the AI recommendation. It is the complement of final AI alignment.

Shift toward AI. A shift toward AI occurs when the participant’s initial judgment differs from the AI recommendation but their final judgment matches the AI recommendation. This measure captures adoption of AI advice within the relevant risk set. We operationalize harmful advice adoption as a change from an initial judgment that disagrees with an incorrect AI recommendation to a final judgment that matches that recommendation.

Shift away from AI. A shift away from AI occurs when the participant’s initial judgment matches the AI recommendation but their final judgment no longer matches it. This measure captures resistance or movement away from AI advice.

Correction and deterioration. A correction occurs when a participant moves from an initially incorrect judgment to a finally correct judgment. A deterioration occurs when a participant moves from an initially correct judgment to a finally incorrect judgment. These measures indicate whether judgment updating improves or harms accuracy.

Confidence change. We calculate confidence change as final confidence minus initial confidence. Positive values indicate increased confidence after the interface stage, while negative values indicate reduced confidence.

Chosen-answer Brier score. We assess confidence–correctness correspondence with a chosen-answer Brier score, adapted from the quadratic scoring rule for probabilistic predictions (Brier, 1950). Because we elicit confidence in the chosen answer rather than a direct probability that the item is TRUE, we do not interpret the score as a standard truth-probability Brier score. We scale final confidence ratings from 0–100 to 0–1 and interpret them as the probability assigned to the participant’s own final answer. For example, if a participant chooses TRUE with 80% confidence, this corresponds to $p(\text{TRUE}) = 0.80$. If the participant chooses FAKE with 80% confidence, this corresponds to $p(\text{TRUE}) = 0.20$. With $y = 1$ for TRUE items and $y = 0$ for FAKE items, we write the score as:

$$\text{Brier}_{ij} = (p_{ij}(\text{TRUE}) - y_{ij})^2,$$

where $p_{ij}(\text{TRUE})$ is participant i ’s confidence-derived probability that item j is true, and y_{ij} is the ground-truth truth label. This formulation is mathematically equivalent to scoring confidence in the chosen answer against final-response correctness. Lower chosen-answer Brier scores indicate better confidence–correctness correspondence. However, this score jointly reflects final accuracy and confidence extremity, so we do not interpret it as a pure calibration measure. We use it alongside accuracy and descriptive reliability analyses to assess whether confidence corresponds to response correctness.

3.10 Statistical analysis plan

We first report descriptive trial-level results across all 70 sessions: initial and final accuracy, judgment-shift rates, confidence change, final AI alignment, override rates, and chosen-answer Brier scores. For AI-present trials, we separate AI-correct from AI-incorrect trials because alignment has opposite

Table 2: Summary of trial-level statistical models.

Model	Trials	Dependent variable	Main predictors	Purpose
M1	C0–C4	Final correctness	Condition, initial correctness, initial confidence	Compare final correctness in each condition with the no-AI baseline.
M2	C1–C4	Final correctness	Condition, AI correctness, condition $\times$ AI correctness, initial correctness, initial confidence	Test whether explanation conditions differ from prediction-only advice in final correctness.
M3	C1–C4	Final AI alignment	Condition, AI correctness, condition $\times$ AI correctness, initial alignment, initial confidence	Test whether participants end with the AI recommendation, separately considering AI-correct and AI-incorrect trials.
M4	C1–C4	Chosen-answer Brier score	Condition, AI correctness, condition $\times$ AI correctness, initial chosen-answer Brier score	Test whether confidence corresponds better to correctness across interface conditions.
M5	Initial non-alignment trials in C1–C4	Advice adoption	Condition, AI correctness, condition $\times$ AI correctness, initial confidence	Test whether participants move toward the AI recommendation when they initially disagreed with it.

implications across these two strata. C0 appears in performance and stability analyses, but not in reliance or advice-adoption analyses because participants in C0 do not receive AI advice. Process logs record interface-stage duration as the elapsed time between entering the interface/advice stage and entering the final-judgment stage. We use this timing variable only as an interpretive sensitivity check, not as a primary outcome.

Mixed-effects models provide the principal exploratory inference. Table 2 summarizes the trials included, dependent variable, main predictors, and purpose of each model. The main independent variable is interface condition. In AI-present models, we also include AI correctness and the condition-by-AI-correctness terms because agreement with AI advice has different implications when the AI is correct versus incorrect. All models include crossed random intercepts for participant and item to account for repeated observations from the same participants and repeated use of the same items.

We use Bayesian logistic mixed-effects models with a logit link for binary outcomes: final correctness (M1 and M2), final AI alignment (M3), and advice adoption (M5). We use a linear mixed-effects model for the chosen-answer Brier score (M4) because it is a continuous trial-level score bounded between 0 and 1. Analyses were conducted in Python 3.9.6 using pandas 2.3.3, NumPy 1.26.4, and statsmodels 0.14.6. The Bayesian logistic mixed-effects models were fitted with statsmodels’ `BinomialBayesMixedGLM` using variational Bayes estimation (`fit_vb`). Fixed-effect coefficients are reported as posterior means, and approximate 95% posterior intervals were computed as posterior mean $\pm$ 1.96 posterior standard deviations from the variational posterior. Odds ratios were obtained by exponentiating the log-odds coefficients and interval limits. The chosen-answer Brier score model was fitted with statsmodels `MixedLM` using maximum likelihood.

After fitting the models, we compare adjacent interface conditions. These comparisons ask whether each added interface feature changes the outcome relative to the immediately simpler condition: C2 versus C1 for adding static SHAP highlights to prediction-only advice, C3 versus C2 for revealing the same highlights sequentially rather than all at once, and C4 versus C3 for adding the boundary cue to sequential reveal. When a model includes AI correctness, we report the same comparison separately for trials where the AI recommendation is correct and where it is incorrect. We report interval estimates as the main inferential evidence and Holm-adjusted approximate two-sided p -values as supplemental summaries. For Bayesian logistic models, intervals refer to approximate 95% posterior intervals for odds ratios; for the linear chosen-answer Brier model, intervals refer to 95% Wald confidence intervals for coefficient differences. We also add the displayed model score to the final-alignment model as a sensitivity check because that score varies at the item level.

4 Results

We organize the results around the research questions. We first describe the final matched sample and data-quality checks, then report overall performance and baseline judgment stability. We then examine judgment change across interface conditions (RQ1), correctness-conditional advice use (RQ2), and confidence change with the chosen-answer Brier score (RQ3). Descriptive trial-level results show the behavioral patterns. Exploratory mixed-effects models provide the principal inferential analyses while accounting for repeated observations from participants and items.

4.1 Participant sample and data quality

The final matched dataset includes 70 participants and 1,750 trial-level observations. Each participant contributed 25 complete trials, with five trials in each interface condition. All 70 matched sessions were retained in the primary analyses, leaving a final sample of $N = 70$.

Six sessions took less than eight minutes and were flagged as potentially fast responses. Excluding these sessions did not materially change the main descriptive patterns, so they were retained in the primary analyses.

The final sample is mostly young adult and university-educated, with high self-reported English reading comfort. These characteristics matter for external validity because responses to AI explanations and confidence scales may differ in broader populations. Appendix A reports participant background characteristics in Table 11.

4.2 Overall performance and baseline judgment stability

Overall, final judgments become less accurate while confidence increases. Across all conditions, initial accuracy is 74.4%, while final accuracy is 71.8%, a decrease of 2.6 percentage points. The transition counts in Table 3 show why this occurs: 167 trials move from an initially correct to a finally incorrect judgment, whereas 121 trials move from an initially incorrect to a finally correct judgment; 1135 stay correct and 327 stay incorrect. A McNemar test comparing initial and final correctness indicates that this deterioration–correction imbalance is statistically significant, $\chi^2(1) = 7.03$, $p = .008$. Because this test is applied to aggregate paired trial transitions and does not model repeated observations from participants and items, we treat it as a descriptive significance check and rely on the mixed-effects models below for the main exploratory inference.

At the same time, mean confidence increases by 6.06 points from initial to final judgment. This means that participants become more confident overall even though final accuracy does not improve. The chosen-answer Brier score changes only slightly, from 0.222 at the initial judgment stage to 0.221 at the final judgment stage, so the aggregate confidence increase does not translate into a clear aggregate improvement in confidence–correctness correspondence.

In aggregate, the interface does not simply improve accuracy. Instead, the second judgment stage is associated with lower final accuracy, higher confidence, and outcome-dependent judgment revision. The analyses below therefore focus on decision dynamics and confidence–correctness correspondence by interface condition and AI correctness.

Table 4 summarizes performance by interface condition. In this table, shift is calculated as the percentage of trials within a condition on which the final judgment differs from the initial judgment, regardless of direction, correctness, or AI alignment. The no-AI baseline condition (C0) has the highest final accuracy at 73.4%; participants’ unaided judgments are therefore already relatively strong in this item set. Among

Table 3: Overall initial-to-final judgment transitions.

Transition type	Count	Percentage of all trials
Initial correct $\rightarrow$ final correct	1135	64.9%
Initial incorrect $\rightarrow$ final incorrect	327	18.7%
Initial incorrect $\rightarrow$ final correct	121	6.9%
Initial correct $\rightarrow$ final incorrect	167	9.5%
Total	1,750	100.0%

Table 4: Judgment performance by interface condition.

Condition	N	Initial acc.	Final acc.	Shift	$\Delta\text{Conf.}$	Chosen-answer Brier
C0 No AI	350	74.3%	73.4%	3.1%	+2.23	0.207
C1 Prediction-only	350	74.9%	70.6%	21.1%	+5.09	0.245
C2 Static SHAP	350	75.4%	70.0%	20.9%	+8.12	0.232
C3 Sequential reveal	350	74.0%	73.1%	17.4%	+10.27	0.215
C4 Sequential + boundary cue	350	73.4%	71.7%	20.0%	+4.62	0.208

the AI-present conditions, C1 and C2 show the clearest descriptive decline from initial to final accuracy. In C1, accuracy decreases from 74.9% to 70.6%; in C2, it decreases from 75.4% to 70.0%. These two conditions also produce relatively high judgment-shift rates, at 21.1% and 20.9% respectively.

C3 and C4 show different patterns. C3 has a smaller decline than C1 and C2, moving from 74.0% to 73.1% and remaining closer to baseline. C4 has lower final accuracy, at 71.7%, but has the lowest final chosen-answer Brier score among AI-present conditions. Accuracy and the chosen-answer Brier score therefore do not point to the same condition-level ranking.

Confidence increases in every condition. The largest increase occurs in C3, where confidence rises by 10.27 points on average. C2 shows a larger descriptive confidence increase than C1, despite similar or lower final accuracy.

The condition-level results do not support a simple claim that more explanation improves performance. Prediction-only advice and static SHAP both increase judgment movement while reducing final accuracy. Sequential reveal and the boundary cue show different profiles. C3 has more favorable final accuracy, whereas C4 has a more favorable chosen-answer Brier score.

4.3 RQ1: Judgment change across interface conditions

To examine reliance on AI advice more directly, the next analysis focuses only on AI-present trials (C1–C4; $N = 1,400$). Table 5 reports initial alignment, final alignment, and the direction of judgment shifts relative to the AI recommendation.

The most stable pattern concerns the direction of judgment revision. C1 and C2 show almost no away-from-AI shifts (0.3% and 0.0%), while C3 and C4 show small away-from-AI rates (0.9% and 1.4%). Most judgment changes move toward the AI recommendation, which is consistent with a strong advice effect.

Final alignment is highest in C3, at 80.0%, followed by C1 and C2 at 79.1%, and C4 at 73.4%. Higher alignment does not imply better performance by itself, because alignment is beneficial only when the AI is correct. High alignment is therefore ambiguous: it may reflect useful reliance when the AI is correct, but it may also reflect incorrect-AI alignment when the AI is wrong.

Table 5: AI alignment and direction of judgment change in AI-present conditions.

Condition	AI correct	Initial align	Final align	Toward AI	Away from AI
C1 Prediction-only	60.0%	58.6%	79.1%	20.9%	0.3%
C2 Static SHAP	60.0%	58.3%	79.1%	20.9%	0.0%
C3 Sequential reveal	60.0%	64.3%	80.0%	16.6%	0.9%
C4 Sequential + boundary cue	60.0%	56.3%	73.4%	18.6%	1.4%

Table 6: AI alignment and confidence outcomes by AI correctness.

AI correctness	<i>N</i>	Final align	Final acc.	Shift	Toward AI	Δ Conf.
AI wrong	560	58.2%	41.8%	30.0%	28.6%	+1.33
AI correct	840	91.1%	91.1%	13.1%	13.0%	+10.81

These results clarify why final accuracy alone is insufficient. A condition can produce high final alignment because final judgments match correct AI advice. It can also produce high alignment because final judgments match incorrect AI advice. The next analysis therefore separates AI-correct and AI-incorrect trials.

Taken together, AI-present interfaces increase judgment movement, but this movement is strongly asymmetric: participants mostly move toward rather than away from AI advice.

4.4 RQ2: Correctness-conditional advice use

Table 6 separates AI-present trials according to whether the AI recommendation is correct. Final alignment is descriptively higher on AI-correct than AI-wrong trials. When the AI is correct, participants align with it on 91.1% of trials. When the AI is wrong, final alignment falls to 58.2%. Because AI correctness is an item-level property rather than a randomized manipulation, this difference may also reflect item difficulty.

The two rates differ descriptively. However, the AI-wrong alignment rate remains high. On trials where the AI recommendation is incorrect, participants still end with the AI-consistent answer in more than half of cases. This is high final alignment with incorrect AI advice. It does not by itself show that AI exposure causes the final AI-aligned answer.

The confidence pattern adds a second layer to this interpretation. On AI-correct trials, confidence increases by 10.81 points on average. On AI-wrong trials, confidence increases by 1.33 points. The average confidence increase is therefore much smaller on AI-wrong trials than on AI-correct trials. Nevertheless, final choices still often move into alignment with incorrect AI recommendations. In other words, the problematic pattern emerges more strongly at the level of judgment direction than at the level of confidence inflation.

These results show that final alignment differs by AI correctness in aggregate, while many final judgments still match incorrect AI recommendations. This descriptive contrast requires interpretation alongside item-level differences and the planned condition contrasts below.

Interface format under AI error. The main research question concerns whether explanation format changes advice use differently when the AI is correct versus incorrect. Table 7 reports condition-level outcomes separated by AI correctness.

The highest rates of incorrect-AI alignment and harmful movement appear in C1 and C2. When the AI

Table 7: Interface-condition outcomes by AI correctness.

Condition	AI	N	Final align	Final acc.	Toward AI	Chosen-answer Brier
C1 Prediction-only	Wrong	140	60.7% (85/140)	39.3% (55/140)	32.1% (45/140)	0.407
C1 Prediction-only	Correct	210	91.4% (192/210)	91.4% (192/210)	13.3% (28/210)	0.136
C2 Static SHAP	Wrong	140	61.4% (86/140)	38.6% (54/140)	32.9% (46/140)	0.413
C2 Static SHAP	Correct	210	91.0% (191/210)	91.0% (191/210)	12.9% (27/210)	0.111
C3 Sequential reveal	Wrong	140	58.6% (82/140)	41.4% (58/140)	22.1% (31/140)	0.399
C3 Sequential reveal	Correct	210	94.3% (198/210)	94.3% (198/210)	12.9% (27/210)	0.092
C4 Sequential + boundary cue	Wrong	140	52.1% (73/140)	47.9% (67/140)	27.1% (38/140)	0.329
C4 Sequential + boundary cue	Correct	210	87.6% (184/210)	87.6% (184/210)	12.9% (27/210)	0.128

Note. Percentages for final alignment, final accuracy, and toward-AI shifts include event counts in parentheses.

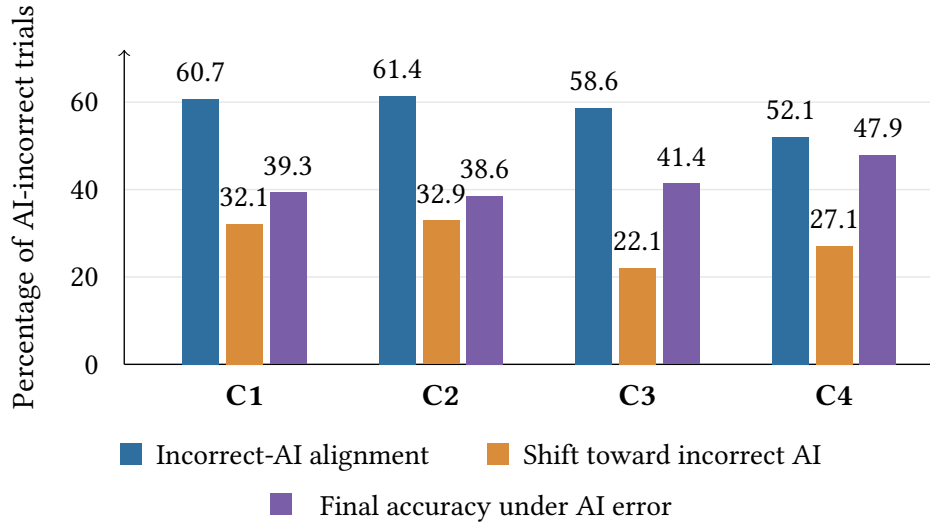


Figure 8: Descriptive outcomes on AI-incorrect trials. C3 has the lowest shift-toward-incorrect-AI rate; C4 has the lowest final incorrect-AI alignment and the highest final accuracy.

is wrong, prediction-only advice produces a final alignment rate of 60.7% (85/140) and a toward-AI shift rate of 32.1% (45/140). Static SHAP explanations produce a very similar pattern, with final alignment at 61.4% (86/140) and toward-AI shift at 32.9% (46/140) under AI error. These one-trial differences are negligible. Thus, static explanations do not show a clear protective advantage over prediction-only advice, but they also do not provide clear evidence of greater harmful advice adoption.

C3 and C4 show comparatively favorable descriptive values on different indicators. C3 has the lowest toward-AI shift rate under AI error, at 22.1% (31/140), compared with 32.9% (46/140) in C2. This gives a 15-trial difference within the AI-wrong condition cells. C4 has the lowest final alignment with incorrect AI advice, at 52.1% (73/140), compared with 58.6% (82/140) in C3. It also has the lowest descriptive chosen-answer Brier score under AI error, at 0.329. These descriptive differences do not show whether the pattern comes from greater awareness of model fallibility, generalized caution, lower trust in the interface, longer exposure, or item-level variation.

The indicators do not identify the same pattern. C3 has the most favorable value for harmful shifts toward incorrect AI advice, whereas C4 has the most favorable values for final incorrect-AI alignment and the descriptive chosen-answer Brier score. These are related but not identical outcomes, and they do not show that the two interfaces operate through distinct psychological mechanisms.

Together, Table 7 and Figure 8 show that the same interface can look comparatively favorable on one outcome but not on another. C3 has the most favorable value for harmful shifts toward incorrect AI

advice, whereas C4 has the most favorable values for final incorrect-AI alignment and the descriptive chosen-answer Brier score. The Discussion returns to this outcome-dependence.

The key result is not that one explanation format clearly wins, but that different reliance metrics lead to different rankings.

Correction and deterioration by condition. To further characterize the cost-benefit profile of each interface, Figure 9 reports corrections and deteriorations by condition. A correction refers to a trial where the participant moves from an initially incorrect judgment to a finally correct judgment. A deterioration refers to the opposite pattern: initially correct to finally incorrect.

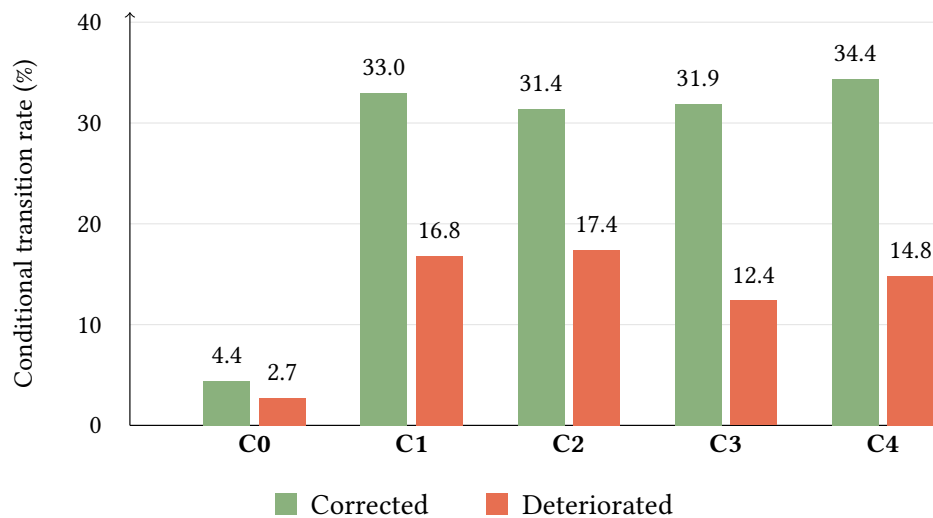


Figure 9: Conditional correction and deterioration rates by interface condition. Correction rate is $P(\text{final correct} \mid \text{initial incorrect})$, and deterioration rate is $P(\text{final incorrect} \mid \text{initial correct})$. Counts for corrections/deteriorations are C0: 4/7, C1: 29/44, C2: 27/46, C3: 29/32, and C4: 32/38.

In absolute counts, deteriorations exceed corrections in every condition. C1 has 29 corrections and 44 deteriorations; C2 has 27 corrections and 46 deteriorations. C3 and C4 show smaller absolute gaps, with 29 corrections versus 32 deteriorations in C3 and 32 corrections versus 38 deteriorations in C4. By contrast, C0 has few changes overall, as expected, because participants do not receive AI advice.

Because corrections and deteriorations have different risk sets, Figure 9 reports conditional rates. The correction rate is the percentage of initially incorrect trials that became correct. The deterioration rate is the percentage of initially correct trials that became incorrect.

The transition analysis shows that the cost-benefit profile depends on the starting judgment. Raw counts describe the overall trial burden, whereas conditional rates show how often initially incorrect judgments become correct and initially correct judgments deteriorate. C1 and C2 have the highest conditional deterioration rates. C3 and C4 have lower deterioration rates while maintaining similar or higher correction rates. These patterns describe asymmetric updating, but they require interpretation alongside initial-state denominators, alignment, and chosen-answer Brier scores rather than as stand-alone evidence of a harmful mechanism.

Table 8: Confidence change and chosen-answer Brier score by interface condition.

Condition	$\Delta\text{Conf.}$	Overall Brier	Brier, AI wrong	Brier, AI correct
C0 No AI	+2.23	0.207	–	–
C1 Prediction-only	+5.09	0.245	0.407	0.136
C2 Static SHAP	+8.12	0.232	0.413	0.111
C3 Sequential reveal	+10.27	0.215	0.399	0.092
C4 Sequential + boundary cue	+4.62	0.208	0.329	0.128

Note. Lower chosen-answer Brier scores indicate better confidence–correctness correspondence for the selected answer. C0 has no AI-wrong or AI-correct strata because participants did not receive AI advice.

4.5 RQ3: Confidence and chosen-answer Brier score

Confidence increased in every condition, but the increase did not track final accuracy. C3 produced the largest mean confidence increase, at +10.27 points, while C4 produced a smaller confidence increase, at +4.62 points. C2 also increased confidence more than C1 (+8.12 versus +5.09) even though C2 did not produce higher final accuracy. This pattern shows why confidence change alone is not a sufficient indicator of improved AI-assisted judgment.

Table 8 summarizes the confidence and chosen-answer Brier outcomes used for RQ3. Overall chosen-answer Brier scores were lowest in C0 and C4 (0.207 and 0.208), followed by C3 (0.215), C2 (0.232), and C1 (0.245). Among AI-present conditions, C4 had the lowest descriptive Brier score when the AI was wrong (0.329), whereas C3 had the lowest descriptive Brier score when the AI was correct (0.092). These patterns do not give the same ranking as final accuracy or advice adoption.

Because the chosen-answer Brier score jointly reflects final accuracy and confidence extremity, we interpret it as confidence–correctness correspondence rather than pure calibration. The RQ3 pattern therefore supports the broader conclusion that explanation formats should be evaluated across multiple outcomes: higher confidence, higher accuracy, lower incorrect-AI alignment, and better confidence–correctness correspondence do not necessarily move together.

4.6 Hypothesis-oriented model summary and sensitivity analyses

Table 9 summarizes the planned contrasts. It separates descriptive patterns from model-estimated contrasts and marks unsupported or inconclusive results explicitly. The adoption model (M5) is especially relevant for harmful movement toward AI because it excludes trials where participants already agreed with the AI before exposure.

Table 9: Hypothesis-oriented decision table.

Hyp.	Primary estimand	Estimate	Interval	Interpretation
H1a	Judgment shift in AI-present conditions versus C0	19.9% (278/1400) vs. 3.1% (11/350)	Descriptive summary	Descriptive only; planned models focus on reliance/adoption.
H1b	Direction of AI-present judgment shifts	Toward AI: 19.2% (269/1400); away: 0.6% (9/1400)	Descriptive summary	Descriptive only; planned models focus on reliance/adoption.
H2	C2 versus C1 final AI alignment	79.1% vs. 79.1%	Descriptive summary	Not clearly supported.
H3	C2 versus C1 final alignment under AI error (M3)	OR = 1.04	[0.72, 1.50]	Not clearly supported.
H4	C3 versus C2 harmful advice adoption under AI error (M5)	OR = 0.71	[0.40, 1.27]	Consistent with expectation, but inconclusive.
H5	C4 versus C3 final incorrect-AI alignment (M3)	OR = 0.93	[0.54, 1.61]	Not clearly supported.
H5	C4 versus C3 beneficial advice adoption when AI is correct (M5)	OR = 0.35	[0.11, 1.12]	Possible trade-off; inconclusive.
H5	C4 versus C3 chosen-answer Brier score under AI error (M4)	$b = -0.036$	[-0.087, 0.014]	Consistent with expectation, but inconclusive.

Note. The table summarizes H1a, H1b, and H2 descriptively because the planned inferential contrasts focus on correctness-conditional reliance and advice adoption. M3 and M5 are Bayesian logistic mixed-effects models; entries report odds ratios with approximate 95% posterior intervals. For odds ratios, values below 1 indicate lower odds for the first condition relative to the comparison condition, whereas values above 1 indicate higher odds. Intervals that include 1 do not provide clear evidence of a directional difference. M5 uses only trials where the initial judgment does not match the AI recommendation. M4 is a linear mixed-effects model; entries report coefficient differences with 95% Wald confidence intervals. For M4, intervals that include 0 do not provide clear evidence of a coefficient difference. None of the planned Holm-adjusted contrasts in Appendix E.6 meet the conventional $p < .05$ threshold, so we do not describe these model-based contrasts as statistically significant. Appendix E reports full coefficients for M1, M2, M3, M4, and M5 (Appendix E.1, E.2, E.3, E.4, and E.5), along with the planned contrasts in Appendix E.6.

Appendix Figure 11 visualizes selected coefficients from the exploratory models. The large AI-correctness coefficient should be read cautiously because it is partly structural: when participants align with correct advice, their final response is correct, whereas alignment with incorrect advice produces an incorrect response. Under AI error, static SHAP does not differ clearly from prediction-only advice in final alignment or advice adoption. Sequential reveal shows lower odds of adopting incorrect AI advice than static SHAP in direction (M5: OR = 0.71), but the posterior interval crosses 1. The boundary-cue condition does not show lower odds of incorrect-advice adoption than sequential reveal (M5: OR = 1.14). The direct C4 versus C3 contrast in the trial-level Brier model also does not clearly support its lower descriptive chosen-answer Brier score under AI error (M4: $b = -0.036$, interval crossing 0).

Thus, the results do not support the expectation that static SHAP would increase AI alignment beyond prediction-only advice; instead, the two conditions produce nearly identical descriptive patterns.

The sensitivity checks do not change the main interpretation. A participant-condition aggregated Brier check shows the same C4 direction under AI error, but not a uniformly favorable pattern across AI correctness; Appendix E.7 reports this check. Adding the displayed model score to the final-alignment model leaves the planned condition contrasts almost unchanged, as shown in Appendix E.8. Process logs also show longer interface-stage duration in sequential conditions (median 7.8 seconds in C3 and 8.8 seconds in C4, compared with 2.6 seconds in C1 and 4.0 seconds in C2). Appendix E.9 reports the duration-exclusion sensitivity check, and Appendix E.10 reports exploratory participant-level moderator checks. These timing differences matter for interpretation: C2 versus C3 and C3 versus C4 are not pure format contrasts.

Table 10: Lowest final-accuracy items by internal task ID.

Internal ID	Short description	Truth	AI	Initial acc.	Final acc.
13	Canada’s personal-data rules	FAKE	TRUE	51.4%	27.1%
16	Harris’ father as Marxist economist	TRUE	FAKE	50.0%	28.6%
22	Eclipse image behind CN Tower	FAKE	TRUE	64.3%	38.6%
20	Canadian tourist-pass conspiracy	FAKE	TRUE	71.4%	44.3%
21	Blue-ringed octopus venom	TRUE	FAKE	67.1%	44.3%
0	The Orbanisation of America	TRUE	FAKE	70.0%	47.1%
12	Hurricane Helene geoengineering claim	FAKE	TRUE	72.9%	52.9%
6	Covid vaccines and cancer claim	FAKE	TRUE	74.3%	57.1%

Note. Internal IDs follow the archived task records and match the ordering in Appendix D.

4.7 Item heterogeneity and sensitivity analyses

Finally, we inspect item-level results to identify whether reliance patterns concentrate in particular stimulus types. Table 10 lists the items with the lowest final accuracy. These items do not automatically indicate invalid stimuli. Rather, they indicate which types of claims are most susceptible to AI-driven updating.

Several high-susceptibility items share common features. Some are counterintuitive true claims, such as the blue-ringed octopus item, where the correct answer is true but the claim sounds exaggerated. Others are politically or semantically loaded, such as the item describing Kamala Harris’ father as a Marxist economist. Still others involve complex institutional or evidential contexts, such as Canadian personal-data rules or an eclipse image circulated as a real photograph. These item types are precisely the kinds of cases in which participants may lack stable prior knowledge and therefore give more weight to external advice.

The item-level pattern suggests that susceptibility to incorrect AI advice may vary with item characteristics. Several low-accuracy items appear counterintuitive, politically charged, visually dependent, or institutionally complex. Because this analysis identifies these characteristics post hoc and does not independently code or experimentally manipulate them, we treat them as hypotheses for future work rather than established moderators.

5 Discussion

Our results complicate a simple transparency-improves-performance account. In our task, explanation presence is less informative than how an interface relates to judgment movement, final alignment, and confidence-related performance when advice can be right or wrong. This pattern points to an outcome-dependent view of XAI evaluation rather than a single ranking of interface formats.

Four findings organize the interpretation. First, AI advice changes many judgments, but final agreement alone does not prove reliance. Second, static SHAP does not improve resistance to incorrect advice relative to prediction-only output. Third, sequential reveal and the boundary cue look favorable on different outcomes. Fourth, accuracy, advice adoption, final alignment, and confidence-related performance do not give the same answer about explanation quality.

5.1 Principal findings

AI advice carries considerable weight in the second judgment stage. Most judgment changes in AI-present trials move toward the AI recommendation. However, final agreement alone does not identify reliance. Some participants already agree with the AI before seeing the recommendation. For that reason, final alignment needs to be read together with toward-AI shifts, deterioration from initially correct to finally incorrect judgments, and advice adoption among initially non-aligned trials.

AI correctness also matters because alignment has opposite meanings across correctness strata. Alignment with correct advice can support performance; alignment with incorrect advice can harm it. This distinction keeps the interpretation focused on selective reliance rather than global agreement with AI.

Statistical significance aside, we note a practically important divergence between judgment quality and subjective certainty: final judgments become slightly less accurate, while confidence increases. The second judgment stage gives participants another opportunity to reconsider, but in our task it can leave them more certain without making them more correct. This pattern is potentially concerning for AI-assisted misinformation evaluation because higher confidence may make incorrect final judgments harder to revise.

5.2 Prediction-only advice and static SHAP

Prediction-only advice and static SHAP lead to the same interpretation: static token-level attribution cues do not function as a clear safeguard against incorrect AI recommendations in this interface and task context. Under AI error, C1 and C2 show very similar final-alignment and harmful-shift patterns. This does not mean that static SHAP adds harm beyond prediction-only advice. It means that adding a static explanation does not, by itself, make reliance more selective.

This result narrows the claim about static attribution displays. In this task, the static display gives participants more model-derived information, but it does not clearly help them resist incorrect advice. Explanation content therefore needs behavioral evaluation. Researchers should not assume that making model output more visible makes judgment more selective.

5.3 Sequential reveal, boundary cue, and selectivity

Sequential reveal and the boundary cue raise different interpretive possibilities. C3 looks most favorable on harmful movement toward incorrect AI advice. C4 looks more favorable on final incorrect-AI alignment and the chosen-answer Brier score under AI error. We should not treat these patterns as evidence of distinct cognitive mechanisms, because we do not directly measure processing depth, attention, perceived AI authority, understanding of attribution cues, or uptake of the boundary cue.

The C4 pattern matters because it suggests a selectivity trade-off. Lower alignment under AI error is desirable only if the interface preserves alignment with correct AI advice. C4 shows lower final alignment when the AI is wrong, but it also shows lower alignment than C3 when the AI is correct. The model results do not clearly establish this difference as an effect, but the pattern still cautions against treating the boundary cue as straightforwardly beneficial. It may encourage generalized resistance rather than calibrated resistance.

The timing evidence introduces a separate interpretation problem. Sequential conditions require additional reveal interactions and involve longer interface-stage duration. Any C2 versus C3 or C3 versus C4 difference may therefore reflect pacing, interaction effort, or time-on-task rather than presentation sequence alone. The lower harmful-shift rate in C3 could come from staged visual presentation, added

friction, longer deliberation time, or a combination of these factors. Our design cannot separate these explanations.

Together, these findings shift the design interpretation. We should not read C3 and C4 as two increasingly stronger safeguards. They appear to change different parts of the decision process and may introduce different costs. For design, the useful question is whether an interface helps users resist incorrect advice without discouraging appropriate use of correct advice.

5.4 Methodological contribution

Our clearest contribution is methodological. By separating final AI agreement, advice-induced movement, final accuracy, deterioration, confidence change, and the chosen-answer Brier score, we show why these outcomes should not be treated as interchangeable measures of explanation quality or appropriate reliance.

The implication is practical: the same interface can look favorable on one outcome but not another. A single global metric would obscure this outcome dependence and could make a design look beneficial even when it shifts the trade-off rather than improving selectivity.

5.5 Design implications

The findings suggest three cautious design implications. First, designers should not assume that attribution explanations protect users merely because they provide more information. Static token highlights do not show a clear protective advantage over prediction-only advice.

Second, explanation timing warrants further investigation, but future studies should examine it together with interaction time and pacing. Sequential reveal may change how users engage with advice, but this experiment cannot separate presentation sequence from time-on-task and interaction effort. A cleaner follow-up design could compare sequential reveal with a time-matched static interface or a static interface with an equivalent enforced delay. This would help isolate sequence from friction.

Third, researchers should evaluate boundary cues for selectivity, not only for reduced agreement with AI. A cue that lowers alignment with incorrect advice may still be problematic if it also lowers beneficial reliance when the AI is correct. Generic warnings may reduce harmful reliance, but they may also encourage unnecessary skepticism toward useful AI support. Future interfaces may need context-sensitive uncertainty cues rather than broad disclaimers, so that caution increases when model error is likely without discouraging appropriate use of correct advice.

5.6 Limitations

Several limitations guide interpretation. We use short headline-and-summary stimuli rather than full news articles, so readers should understand the task as a controlled veracity-judgment task based on limited text. The sample is moderate: 70 participants complete 1,750 trials, but each participant sees only a small number of AI-incorrect trials per condition. We therefore interpret small condition differences cautiously, especially when model intervals are wide.

The stimulus labels and fixed AI predictions also constrain interpretation. The stimulus pool intentionally includes AI errors, and AI-correct and AI-wrong items may differ in difficulty, truth structure, or topic. The item-level analysis suggests that low-accuracy items often appear counterintuitive, politically charged, visually dependent, or institutionally complex, but these characteristics come from post hoc interpretation. Future work should treat them as hypotheses rather than established moderators.

We do not directly measure the proposed cognitive mechanisms. We also do not include manipulation checks for whether participants notice, understand, or use the SHAP highlights. Some participants may have treated the highlighted tokens as decorative emphasis, ignored them, or misunderstood what the colors and reveal sequence indicated. This limits interpretation of the explanation-format comparisons: observed differences cannot show that participants processed the attribution cues as intended. The same issue applies to the boundary cue, which participants may not have read closely or interpreted as a reminder of model fallibility. The analysis therefore treats SHAP highlights and the boundary cue as interface cues presented to participants, not as verified inputs to participants' reasoning.

Finally, the outcome measures have limits. The binary TRUE/FAKE response captures categorical judgment revision rather than continuous belief updating. The task measures confidence for the chosen answer, not as a direct probability that the claim is true. The chosen-answer Brier score therefore reflects confidence–correctness correspondence for the selected response, so we do not treat it as a pure calibration measure.

5.7 Future work

Future work should build on our design by testing explanation formats under stronger and more realistic conditions. Larger and more diverse samples would make it possible to estimate responses to AI errors more precisely and to examine whether selective reliance differs across user groups. Richer news materials, including longer articles, source cues, images, or platform context, would also help test whether the observed judgment-updating patterns extend beyond a controlled headline-and-summary task.

A second direction is to study explanation processing more directly. Follow-up studies could add manipulation checks, attention measures, or post-task comprehension questions to assess whether users notice and understand SHAP highlights, sequential reveal steps, and boundary cues. This would help distinguish interface exposure from actual explanation use.

Finally, future work could compare token-level SHAP explanations with other forms of AI support, such as example-based explanations, uncertainty displays, model-level accuracy information, explicit warnings about possible AI error, or generative-AI explanations that interpret evidence in an audience-targeted narrative. Such systems may make explanations easier to read, but they could also introduce persuasive framing or unsupported rationales. These comparisons would clarify whether the present findings are specific to attribution-based explanations or reflect broader patterns in AI-assisted misinformation evaluation.

5.8 Implementation, reproducibility, and ethics

We implemented the experiment with a custom online interface and fixed item-level AI advice. The appendix reports the questionnaire items, interface screenshots, stimulus provenance, and model summaries, and the Procedure section links the public interface demonstration. The code and data used for the analyses are available in a public GitHub repository.⁶

XAI also has an ethical role because explanations can shape trust, resistance, and confidence in automated advice. In misinformation judgment, an explanation may be technically faithful while still encouraging agreement with an incorrect recommendation. We therefore treat XAI as a user-facing intervention and use the study to test whether explanation formats support calibrated reliance, complementing model-centered XAI evaluation with evidence about decisions, confidence, and exposure to AI error.

⁶<https://github.com/itsyuning/xai-ui-study>

The procedure includes informed consent before data collection and a debrief after the task, as described in Section 3.1. We collect no names, email addresses, visual data, audio data, biometric data, or other directly identifying information. Because the task involves potentially misleading news-style items and fallible AI advice, the debrief explains why the study includes misleading content and imperfect AI advice, offers optional item clarification, and provides a downloadable or savable debrief record.

6 Conclusion

We examine how AI advice and token-level explanation presentation relate to judgment updating, alignment, confidence change, and confidence–correctness correspondence in a controlled misinformation-evaluation task. The results do not show a simple overall accuracy benefit of AI assistance, nor do they show that static SHAP improves resistance to incorrect advice relative to prediction-only output.

The main message is that explanation formats produce outcome-dependent profiles. Sequential reveal and the boundary cue look comparatively favorable on different measures, but the exploratory analyses do not establish that either format improves selective reliance. Final AI agreement, advice-induced movement, accuracy, deterioration, and confidence-related performance therefore need to remain separate outcomes rather than interchangeable measures of explanation quality.

These findings are bounded by our model, SHAP presentation, stimulus set, participant sample, and online task. Future work should use larger samples, direct measures of proposed cognitive mechanisms, and richer item-level controls before making broader recommendations about explanation timing or boundary cues. Taken together, our results suggest that XAI evaluation should ask not only whether explanations increase agreement with AI advice, but whether they help users update in the right direction when AI advice is correct and resist it when it is wrong.

Appendix overview

The appendices provide supporting material for the Method and Results sections. Appendix A describes the final participant sample. Appendix B lists the background questionnaire items. Appendix C shows the shared task-flow screens. Appendix D reports stimulus provenance and fixed AI predictions. Appendix E provides the exploratory mixed-effects model details, planned contrasts, and sensitivity checks.

A Participant background characteristics

Table 11 summarizes the background characteristics of the final matched participant sample.

Table 11: Participant background characteristics.

Characteristic	Summary
Final matched sample	$N = 70$ participants
Age group	18–24: 25.7%; 25–34: 38.6%; 35–44: 22.9%; 45–54: 7.1%; 55–64: 5.7%.
Gender	Female: 50.0%; Male: 47.1%; Non-binary / third gender: 2.9%.
Education level	Bachelor’s: 52.9%; Master’s: 24.3%; other or lower levels: 21.4%; prefer not to say: 1.4%.
English reading comfort	$M = 6.57$, $SD = 0.86$ on a 7-point scale.
Online news-reading frequency	At least 3 days per week: 74.3%; less often: 25.7%.
Familiarity with misinformation/fact-checking	$M = 5.13$, $SD = 1.32$ on a 7-point scale.
Caution toward AI-generated outputs	$M = 5.46$, $SD = 1.20$ on a 7-point scale.

B Full background questionnaire items

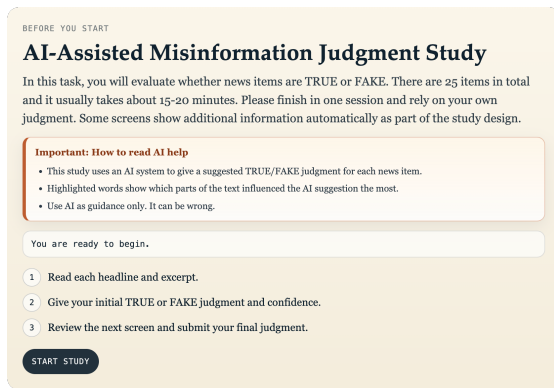
This appendix reproduces the participant background questionnaire items from the study, excluding the consent item.

Variable	Question wording	Response format
Age	How old are you?	Single choice: 18–24 years old; 25–34 years old; 35–44 years old; 45–54 years old; 55–64 years old.
Gender	How do you describe yourself?	Single choice: Male; Female; Non-binary / third gender; optional self-describe free-text field.
Culture	List of Countries	Single choice from country list.
First language	What is your first or primary language?	Open text.

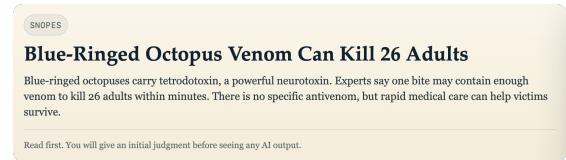
Variable	Question wording	Response format
Education	What is the highest level of education you have completed?	Single choice: Secondary; Vocational or Similar; Some University but no degree; University - Bachelors Degree; Master's degree (MA, MSc, MPhil, etc.); Prefer not to say.
English comfort	How comfortable are you reading short English news texts?	7-point scale; higher values indicate greater comfort (labeled points in export include 4 = Moderately comfortable, 7 = Very comfortable).
English reading ability	How would you rate your English reading ability?	7-point scale; higher values indicate stronger ability (labeled points in export include 4 = Moderate, 7 = Very strong).
News frequency	How often do you read news online?	Single choice: Never; Less than once a week; 1 to 2 days a week; 3 to 6 days a week; Every day.
Misinformation familiarity	How familiar are you with misinformation, fake news, and fact-checking practices?	7-point scale: 1 = Not at all familiar; 4 = Moderately familiar; 7 = Very familiar.
AI usefulness	How helpful do you generally find AI-generated predictions or explanations when making judgments?	7-point scale: 1 = Not at all helpful; 4 = Moderately helpful; 7 = Very helpful.
AI verification tendency	When AI-generated predictions or explanations are provided, how likely are you to verify them before relying on them?	7-point scale: 1 = Very unlikely; 4 = Moderately likely; 7 = Very likely.
AI caution	How cautious are you when using AI-generated outputs to make judgments?	7-point scale; higher values indicate greater caution (labeled points in export include 4 = Moderately cautious, 7 = Very cautious).

C Task-flow interface screenshots

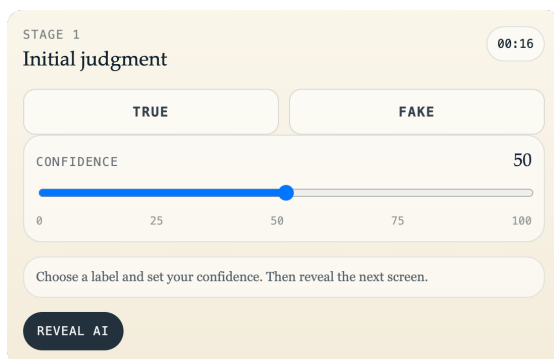
Figure 10 documents the shared task-flow screens that participants saw before and after the condition-specific interface stage.



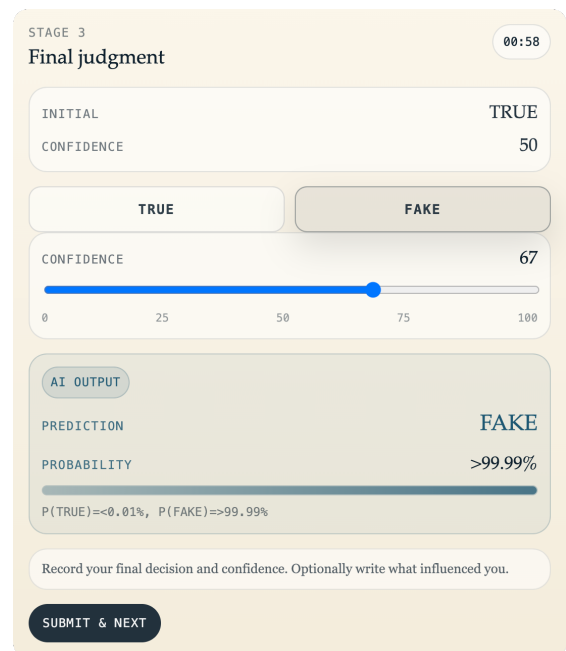
(a) Instruction screen



(b) Stimulus reading view



(c) Initial judgment stage



(d) Final judgment stage

Figure 10: Task-flow interface screenshots for the instruction, item reading, initial judgment, and final judgment stages.

D Stimulus items and AI predictions

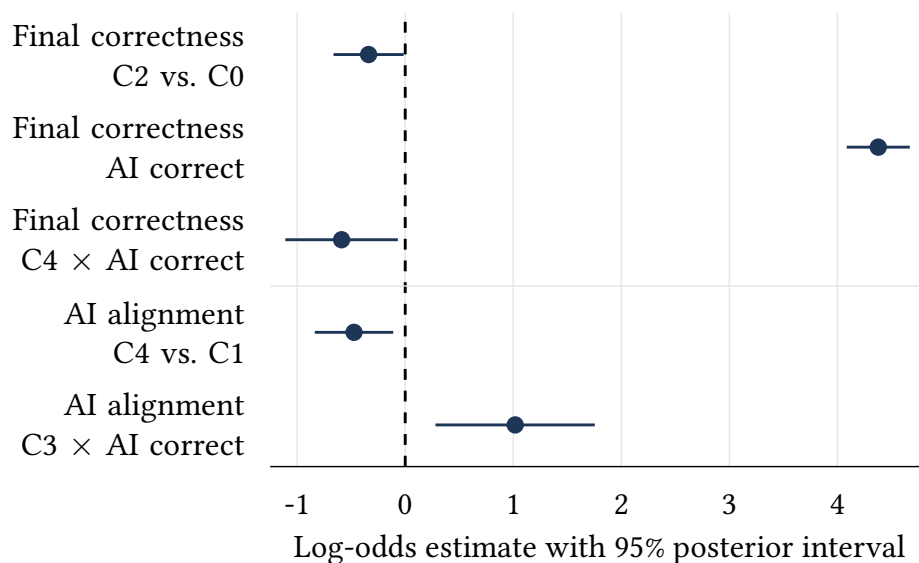
Table 13 lists the 25 stimulus items in the experiment, ordered by internal task ID. The table includes topic category, ground-truth label, AI prediction, AI correctness, and source provenance.

Table 13: Stimulus provenance and AI correctness by internal task ID.

Internal ID	Topic	Truth	AI pred.	AI correct	Source/provenance	Title
0	POLITICS	TRUE	FAKE	No	European Council on Foreign Relations	The Orbanisation of America: Hungary's lessons for Donald Trump
1	POLITICS	TRUE	TRUE	Yes	Center on Budget and Policy Priorities	Congress begins new budget talks as shutdown deadlines approach
2	POLITICS	FAKE	FAKE	Yes	Snopes	Mormon Founder Joseph Smith Was 1st Assassinated US Presidential Candidate
3	POLITICS	FAKE	FAKE	Yes	Snopes	Viral post claims a Facebook copyright disclaimer blocks photo use
4	POLITICS	TRUE	TRUE	Yes	APRI - Africa Policy Research Institute	South Africa coalition talks continue amid disputes over cabinet posts and budget plans
5	HEALTH	TRUE	TRUE	Yes	The World Economic Forum	Researchers report advances across several cancer treatment approaches
6	HEALTH	FAKE	TRUE	No	AFP Fact Check	Covid vaccines linked to study on rising cancer cases
7	HEALTH	TRUE	FAKE	No	University of Bristol	August: COVID-19 and mental illnesses
8	HEALTH	TRUE	TRUE	Yes	Cancer News	NHS expands enrollment pathways for personalized cancer vaccine trials
9	HEALTH	TRUE	TRUE	Yes	Frontiers	German hospitals expand sustainability initiatives in clinical operations
10	SCI_TECH	TRUE	TRUE	Yes	Rice University	Artemis II delay leads NASA to review mission-readiness plans
11	SCI_TECH	TRUE	TRUE	Yes	cio.com	NASA appoints its first chief AI officer
12	SCI_TECH	FAKE	TRUE	No	AFP Fact Check	Hurricane Helene 'geoengineered' to expand lithium mining
13	SCI_TECH	FAKE	TRUE	No	AFP Fact Check	Canada's personal-data rules were quietly expanded
14	SCI_TECH	TRUE	TRUE	Yes	Cybernews	NASA says AI tools could help plan future life-detection missions
15	BUSINESS	FAKE	FAKE	Yes	Snopes	Trump Said He Would Not Defend NATO Allies If Russia Attacks Them

Internal ID	Topic	Truth	AI pred.	AI correct	Source/provenance	Title
16	BUSINESS	TRUE	FAKE	No	Snopes	Kamala Harris' Father Has Been Described as a Marxist Economist
17	BUSINESS	TRUE	TRUE	Yes	Nature	Saudi labour-market study analyses links between inflation, oil prices, trade, and unemployment
18	BUSINESS	TRUE	TRUE	Yes	PolitiFact	Crime is down in Venezuela, not as much as Donald Trump says
19	BUSINESS	TRUE	FAKE	No	Investopedia	What Is Stagflation, What Causes It, and Why Is It Bad?
20	SOCIETY	FAKE	TRUE	No	AFP Fact Check	Posts claim canadian tourist pass reignites surveillance conspiracy theories
21	SOCIETY	TRUE	FAKE	No	Snopes	Blue-Ringed Octopus Venom Can Kill 26 Adults
22	SOCIETY	FAKE	TRUE	No	AFP Fact Check	Viral posts claim eclipse image behind CN Tower was real
23	SOCIETY	FAKE	FAKE	Yes	AFP Fact Check	Posts Claim Australia Listed George Soros as a 'Global Terrorist'
24	SOCIETY	FAKE	FAKE	Yes	AFP Fact Check	Posts Claim Australia Signed WEF Treaty to Close Banks

A. Logistic mixed-effects models



B. Linear mixed-effects model for chosen-answer Brier score

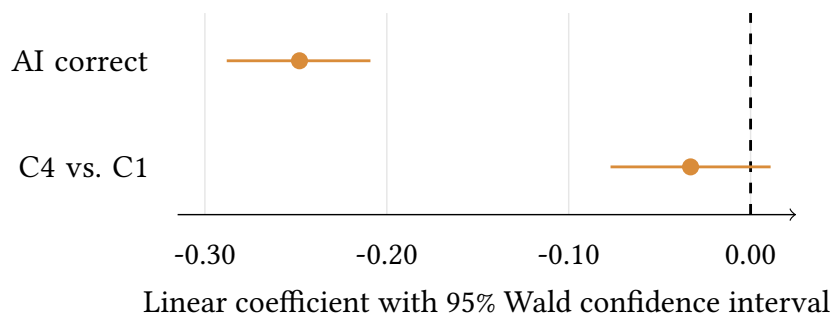


Figure 11: Selected fixed-effect estimates from the exploratory mixed-effects analyses. Points indicate coefficient estimates and horizontal lines indicate 95% intervals. Panel A shows log-odds coefficients with approximate posterior intervals for final correctness and AI alignment. Panel B shows linear coefficients with Wald confidence intervals for chosen-answer Brier score. Negative values in Panel B indicate lower chosen-answer Brier scores and better confidence–correctness correspondence. Dashed vertical lines mark the null value of zero.

E Exploratory mixed-effects model details

This appendix reports full fixed-effect coefficients for the exploratory mixed-effects analyses in the Results section. Figure 11 first gives a compact visual summary of focal coefficients. Appendices E.1–E.5 then report the five model coefficient tables. Appendix E.6 reports the pre-specified pairwise contrasts, and the remaining subsections report robustness, sensitivity, and exploratory moderator checks.

For Bayesian logistic mixed models (M1–M3 and M5), coefficients are posterior means from statsmodels BinomialBayesMixedGLM variational Bayes estimation, with approximate 95% posterior intervals (mean $\pm$ 1.96 posterior SD) and odds ratios (OR). For the linear mixed model (M4), fitted with statsmodels MixedLM, we report coefficients with standard errors, 95% Wald confidence intervals, and p -values.

E.1 M1: Final correctness on all trials

Term	β	95% posterior interval	OR	OR 95% posterior interval
Intercept	-0.975	[-1.125, -0.824]	0.38	[0.32, 0.44]
C1 vs C0	-0.279	[-0.606, 0.048]	0.76	[0.55, 1.05]
C2 vs C0	-0.339	[-0.664, -0.014]	0.71	[0.52, 0.99]
C3 vs C0	0.055	[-0.309, 0.419]	1.06	[0.73, 1.52]
C4 vs C0	-0.139	[-0.468, 0.189]	0.87	[0.63, 1.21]
Initial correctness	3.988	[3.802, 4.175]	53.97	[44.79, 65.02]
Initial confidence (z)	0.223	[0.075, 0.372]	1.25	[1.08, 1.45]

E.2 M2: Final correctness on AI-present trials

Term	β	95% posterior interval	OR	OR 95% posterior interval
Intercept	-3.653	[-3.823, -3.484]	0.03	[0.02, 0.03]
C2 vs C1	-0.161	[-0.492, 0.170]	0.85	[0.61, 1.18]
C3 vs C1	0.255	[-0.115, 0.625]	1.29	[0.89, 1.87]
C4 vs C1	0.292	[-0.034, 0.617]	1.34	[0.97, 1.85]
AI correct	4.379	[4.087, 4.671]	79.74	[59.56, 106.77]
C2 $\times$ AI correct	0.245	[-0.333, 0.824]	1.28	[0.72, 2.28]
C3 $\times$ AI correct	0.244	[-0.432, 0.921]	1.28	[0.65, 2.51]
C4 $\times$ AI correct	-0.589	[-1.110, -0.068]	0.55	[0.33, 0.93]
Initial correctness	4.010	[3.804, 4.216]	55.15	[44.89, 67.75]
Initial confidence (z)	0.219	[0.051, 0.387]	1.24	[1.05, 1.47]

E.3 M3: Final alignment on AI-present trials

Term	β	95% posterior interval	OR	OR 95% posterior interval
Intercept	-0.219	[-0.408, -0.030]	0.80	[0.66, 0.97]
C2 vs C1	0.038	[-0.330, 0.406]	1.04	[0.72, 1.50]
C3 vs C1	-0.402	[-0.816, 0.012]	0.67	[0.44, 1.01]
C4 vs C1	-0.474	[-0.837, -0.111]	0.62	[0.43, 0.89]
AI correct	0.935	[0.610, 1.259]	2.55	[1.84, 3.52]
C2 $\times$ AI correct	0.071	[-0.566, 0.709]	1.07	[0.57, 2.03]
C3 $\times$ AI correct	1.018	[0.280, 1.755]	2.77	[1.32, 5.78]
C4 $\times$ AI correct	0.271	[-0.304, 0.846]	1.31	[0.74, 2.33]
Initial alignment	4.834	[4.187, 5.480]	125.65	[65.83, 239.85]
Initial confidence (z)	-0.598	[-0.789, -0.408]	0.55	[0.45, 0.67]

E.4 M4: Final chosen-answer Brier score on AI-present trials

Term	b	95% Wald confidence interval	SE	p
Intercept	0.253	[0.221, 0.285]	0.016	< .001
C2 vs C1	0.015	[-0.029, 0.058]	0.022	.502
C3 vs C1	0.003	[-0.047, 0.054]	0.026	.899
C4 vs C1	-0.033	[-0.077, 0.011]	0.022	.140
AI correct	-0.248	[-0.288, -0.209]	0.020	< .001
C2 $\times$ AI correct	-0.037	[-0.096, 0.022]	0.030	.217
C3 $\times$ AI correct	-0.054	[-0.119, 0.011]	0.033	.106
C4 $\times$ AI correct	0.021	[-0.039, 0.080]	0.031	.502
Initial chosen-answer Brier	0.601	[0.557, 0.644]	0.022	< .001

E.5 M5: Advice adoption on initial non-alignment trials

Term	β	95% posterior interval	OR	OR 95% posterior interval
Intercept	-0.220	[-0.419, -0.021]	0.80	[0.66, 0.98]
C2 vs C1	-0.008	[-0.395, 0.379]	0.99	[0.67, 1.46]
C3 vs C1	-0.348	[-0.785, 0.088]	0.71	[0.46, 1.09]
C4 vs C1	-0.220	[-0.601, 0.161]	0.80	[0.55, 1.18]
AI correct	0.674	[0.319, 1.028]	1.96	[1.38, 2.80]
C2 $\times$ AI correct	0.169	[-0.522, 0.860]	1.18	[0.59, 2.36]
C3 $\times$ AI correct	1.221	[0.423, 2.018]	3.39	[1.53, 7.52]
C4 $\times$ AI correct	0.034	[-0.601, 0.669]	1.03	[0.55, 1.95]
Initial confidence (z)	-0.649	[-0.850, -0.448]	0.52	[0.43, 0.64]

E.6 Planned contrasts from M3, M4, and M5

Model / outcome	Trials	Contrast	Estimate	95% interval	Holm p
M3 final alignment	AI wrong	C2 vs C1	OR = 1.04	[0.72, 1.50]	1.000
M3 final alignment	AI wrong	C3 vs C2	OR = 0.64	[0.37, 1.12]	.359
M3 final alignment	AI wrong	C4 vs C3	OR = 0.93	[0.54, 1.61]	1.000
M3 final alignment	AI correct	C2 vs C1	OR = 1.12	[0.53, 2.33]	.772
M3 final alignment	AI correct	C3 vs C2	OR = 1.66	[0.54, 5.09]	.751
M3 final alignment	AI correct	C4 vs C3	OR = 0.44	[0.15, 1.31]	.418
M5 advice adoption	AI wrong	C2 vs C1	OR = 0.99	[0.67, 1.46]	1.000
M5 advice adoption	AI wrong	C3 vs C2	OR = 0.71	[0.40, 1.27]	.760
M5 advice adoption	AI wrong	C4 vs C3	OR = 1.14	[0.64, 2.03]	1.000
M5 advice adoption	AI correct	C2 vs C1	OR = 1.17	[0.53, 2.59]	.690
M5 advice adoption	AI correct	C3 vs C2	OR = 2.04	[0.61, 6.80]	.495
M5 advice adoption	AI correct	C4 vs C3	OR = 0.35	[0.11, 1.12]	.231
M4 chosen-answer Brier	AI wrong	C2 vs C1	$b = 0.014$	[-0.036, 0.065]	1.000
M4 chosen-answer Brier	AI wrong	C3 vs C2	$b = -0.012$	[-0.062, 0.038]	1.000
M4 chosen-answer Brier	AI wrong	C4 vs C3	$b = -0.036$	[-0.087, 0.014]	.487
M4 chosen-answer Brier	AI correct	C2 vs C1	$b = -0.022$	[-0.063, 0.019]	.361
M4 chosen-answer Brier	AI correct	C3 vs C2	$b = -0.028$	[-0.069, 0.013]	.361
M4 chosen-answer Brier	AI correct	C4 vs C3	$b = 0.037$	[-0.004, 0.078]	.231

For M3 and M5, intervals are approximate posterior intervals from the variational posterior. For M4, intervals are Wald confidence intervals. Holm-adjusted p -values are approximate two-sided values computed within each model and AI-correctness stratum. Odds ratios below 1 indicate lower odds for the first condition named in the contrast relative to the second condition.

E.7 Participant-aggregated chosen-answer Brier robustness check

Trials	Condition or contrast	Mean or difference	95% interval	<i>p</i>
AI wrong	C1	0.417	[0.350, 0.485]	–
AI wrong	C2	0.437	[0.371, 0.503]	–
AI wrong	C3	0.416	[0.347, 0.486]	–
AI wrong	C4	0.336	[0.269, 0.403]	–
AI wrong	C2 – C1	0.020	[-0.062, 0.101]	.632
AI wrong	C3 – C2	-0.021	[-0.102, 0.060]	.612
AI wrong	C4 – C3	-0.080	[-0.157, -0.004]	.040
AI correct	C1	0.139	[0.104, 0.174]	–
AI correct	C2	0.116	[0.080, 0.153]	–
AI correct	C3	0.096	[0.061, 0.130]	–
AI correct	C4	0.131	[0.095, 0.168]	–
AI correct	C2 – C1	-0.023	[-0.054, 0.009]	.156
AI correct	C3 – C2	-0.021	[-0.055, 0.014]	.232
AI correct	C4 – C3	0.036	[-0.002, 0.074]	.061

Means are participant-condition averages. Intervals and *p*-values for contrasts come from paired participant-level differences ($N = 70$). This robustness check is descriptive and does not replace the trial-level mixed-effects model.

E.8 Sensitivity model with displayed model score

Term	β	95% posterior interval	OR	OR 95% posterior interval
Intercept	-0.136	[-0.325, 0.053]	0.87	[0.72, 1.05]
C2 vs C1	0.020	[-0.348, 0.387]	1.02	[0.71, 1.47]
C3 vs C1	-0.419	[-0.835, -0.003]	0.66	[0.43, 1.00]
C4 vs C1	-0.468	[-0.831, -0.106]	0.63	[0.44, 0.90]
AI correct	0.764	[0.442, 1.087]	2.15	[1.56, 2.97]
C2 $\times$ AI correct	0.068	[-0.567, 0.702]	1.07	[0.57, 2.02]
C3 $\times$ AI correct	1.019	[0.287, 1.751]	2.77	[1.33, 5.76]
C4 $\times$ AI correct	0.257	[-0.315, 0.828]	1.29	[0.73, 2.29]
Displayed model score (z)	0.285	[0.118, 0.452]	1.33	[1.13, 1.57]
Initial alignment	4.844	[4.198, 5.490]	126.94	[66.52, 242.22]
Initial confidence (z)	-0.607	[-0.799, -0.415]	0.54	[0.45, 0.66]

Trials	Contrast	Base M3 OR	With displayed model score OR
AI wrong	C2 vs C1	1.04 [0.72, 1.50]	1.02 [0.71, 1.47]
AI wrong	C3 vs C2	0.64 [0.37, 1.12]	0.65 [0.37, 1.12]
AI wrong	C4 vs C3	0.93 [0.54, 1.61]	0.95 [0.55, 1.65]
AI correct	C2 vs C1	1.12 [0.53, 2.33]	1.09 [0.52, 2.27]
AI correct	C3 vs C2	1.66 [0.54, 5.09]	1.67 [0.55, 5.10]
AI correct	C4 vs C3	0.44 [0.15, 1.31]	0.44 [0.15, 1.31]

We z-standardise the displayed model score across AI-present trials. The covariate is item-level, so we do not interpret it as a causal score effect.

E.9 Sensitivity check (exclude sessions under 8 minutes)

We re-estimate the final-alignment model (M3) after excluding six sessions with duration under 8 minutes. Key terms remain directionally consistent:

- C3 vs C1 (AI-wrong baseline): $\beta = -0.421$ (SD = 0.218)
- C4 vs C1 (AI-wrong baseline): $\beta = -0.434$ (SD = 0.192)
- AI correct: $\beta = 0.933$ (SD = 0.173)
- C3 $\times$ AI correct: $\beta = 0.795$ (SD = 0.399)
- C4 $\times$ AI correct: $\beta = 0.060$ (SD = 0.305)

E.10 Exploratory participant-level moderator checks

As an appendix-only exploratory analysis, we examine whether participant-level survey variables relate to lower alignment with incorrect AI advice. These checks are not part of the primary research question, and we interpret them cautiously given the participant sample size ($N = 70$). For survey items exported with mixed numeric/label strings (e.g., “7 - Very likely”), we use the leading numeric value to recover the 1–7 scale.

First, we estimate a Bayesian logistic mixed model on AI-present trials with participant- and item-level random intercepts:

$$\text{final_align} \sim \text{condition} * \text{ai_correct} + \text{initial_align} + \text{z_initial_conf} \\ + \text{ai_caution_z} + \text{verify_ai_z} + \text{ai_usefulness_z}.$$

Term	β	95% posterior interval	OR	OR 95% posterior interval
AI caution (z)	0.051	[-0.144, 0.247]	1.05	[0.87, 1.28]
Verify AI before relying (z)	-0.184	[-0.379, 0.010]	0.83	[0.68, 1.01]
Perceived AI usefulness (z)	0.109	[-0.088, 0.306]	1.11	[0.92, 1.36]

None of these exploratory participant-level coefficients has a 95% interval excluding 0.

Second, to examine incorrect-AI alignment at the participant level, we compute each participant’s alignment rate on AI-wrong trials (7–9 AI-present trials per participant, depending on list; mean rate = 0.584, $SD = 0.235$), then correlate this rate with selected survey variables using Spearman correlations:

Variable	n	Spearman ρ	p
AI caution	70	-0.072	.556
Likelihood of verifying AI	70	-0.106	.385
AI usefulness	70	0.059	.626
Misinformation familiarity	70	0.030	.807
English reading ability	70	0.144	.235

The exploratory pattern is directionally consistent with the idea that higher caution and verification tendency may relate to lower AI-wrong alignment, but the effects are small and uncertain in this sample.

References

- Abdul, A., Vermeulen, J., Wang, D., Lim, B. Y., & Kankanhalli, M. (2018). Trends and trajectories for explainable, accountable and intelligible systems: An HCI research agenda. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–18. <https://doi.org/10.1145/3173574.3174156>
- Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., & Kim, B. (2018). Sanity checks for saliency maps [NeurIPS 2018]. *Advances in Neural Information Processing Systems*. <https://arxiv.org/abs/1810.03292>
- Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M. T., & Weld, D. (2021). Does the whole exceed its parts? the effect of AI explanations on complementary team performance. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–16. <https://doi.org/10.1145/3411764.3445717>
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3. [https://doi.org/10.1175/1520-0493\(1950\)078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2)
- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 1–21. <https://doi.org/10.1145/3449287>
- Chen, V., Liao, Q. V., Vaughan, J. W., & Bansal, G. (2023). Understanding the role of human intuition on reliance in human-AI decision-making with explanations. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW2), 1–32. <https://doi.org/10.1145/3610219>
- de Jong, S., Bos, M. W., van Berkel, N., & Lamers, M. H. (2025). Algorithm appreciation or aversion: The effects of accuracy disclosure on users’ reliance on algorithmic suggestions. *Behaviour & Information Technology*, 1–20. <https://doi.org/10.1080/0144929X.2025.2535732>
- Delaunay, J., Galarraaga, L., Largouet, C., & van Berkel, N. (2025). Impact of explanation technique and representation on users’ comprehension and confidence in explainable ai. *Proceedings of the ACM on Human-Computer Interaction*, 9(2), 1–28. <https://doi.org/10.1145/3711011>
- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114–126. <https://doi.org/10.1037/xge0000033>
- Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning [arXiv:1702.08608]. *arXiv*. <https://arxiv.org/abs/1702.08608>
- Dzindolet, M. T., Pierce, L. G., Beck, H. P., & Dawe, L. A. (2002). The perceived utility of human and automated aids in a visual detection task. *Human Factors*, 44(1), 79–94. <https://doi.org/10.1518/0018720024494856>
- Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., Kendeou, P., Vraga, E. K., & Amazeen, M. A. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology*, 1(1), 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Gruzd, A., Mai, P., & Soares, F. B. (2024). To share or not to share: Randomized controlled study of misinformation warning labels on social media. *Disinformation in Open Online Media: MISDOOM 2024*, 15175, 46–69. https://doi.org/10.1007/978-3-031-71210-4_4
- Hase, P., & Bansal, M. (2020). Evaluating explainable AI: Which algorithmic explanations help users predict model behavior? *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 5540–5552. <https://doi.org/10.18653/v1/2020.acl-main.491>
- Hashemi Chaleshtori, F., Ghosal, A., Gill, A., Bambroo, P., & Marasovic, A. (2024). On evaluating explanation utility for human-AI decision making in NLP. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 7456–7504. <https://doi.org/10.18653/v1/2024.findings-emnlp.439>

- Hoes, E., Aitken, B., Zhang, J., Gackowski, T., & Wojcieszak, M. (2024). Prominent misinformation interventions reduce misperceptions but increase scepticism. *Nature Human Behaviour*, 8, 1545–1553. <https://doi.org/10.1038/s41562-024-01884-x>
- Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, 57(3), 407–434. <https://doi.org/10.1177/0018720814547570>
- Jacovi, A., & Goldberg, Y. (2020). Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4198–4208. <https://doi.org/10.18653/v1/2020.acl-main.386>
- Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H., & Wortman Vaughan, J. (2020). Interpreting interpretability: Understanding data scientists’ use of interpretability tools for machine learning. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–14.
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). Roberta: A robustly optimized BERT pretraining approach [arXiv:1907.11692]. *arXiv*. <https://arxiv.org/abs/1907.11692>
- Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103. <https://doi.org/10.1016/j.obhdp.2018.12.005>
- Lundberg, S. M., & Lee, S. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. <https://arxiv.org/abs/1705.07874>
- Muralidhar, D. (2024). The effect of progressive disclosure in the transparency of explainable artificial intelligence systems. *Proceedings of the 2024 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 382–383. <https://doi.org/10.1109/VL/HCC60511.2024.00057>
- Nguyen, D. (2018). Comparing automatic and human evaluation of local explanations for text classification. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, 1069–1078. <https://doi.org/10.18653/v1/N18-1097>
- Nyhan, B., & Reifler, J. (2010). When corrections fail: The persistence of political misperceptions. *Political Behavior*, 32, 303–330. <https://doi.org/10.1007/s11109-010-9112-2>
- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230–253. <https://doi.org/10.1518/001872097778543886>
- Pareek, S., van Berkel, N., Velloso, E., & Goncalves, J. (2024). Effect of explanation conceptualisations on trust in ai-assisted credibility assessment. *Proceedings of the ACM on Human-Computer Interaction*, 8(CSCW2), 1–31. <https://doi.org/10.1145/3686922>
- Pennycook, G., Berinsky, A. J., Bhargava, P., Lin, H., Cole, R., Goldberg, B., Lewandowsky, S., & Rand, D. G. (2024). Inoculation and accuracy prompting increase accuracy discernment in combination but not alone. *Nature Human Behaviour*, 8(12), 2330–2341. <https://doi.org/10.1038/s41562-024-02023-2>
- Pennycook, G., & Rand, D. G. (2019). Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition*, 188, 39–50. <https://doi.org/10.1016/j.cognition.2018.06.011>
- Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. *Trends in Cognitive Sciences*, 25(5), 388–402. <https://doi.org/10.1016/j.tics.2021.02.007>
- Poursabzi-Sangdeh, F., Goldstein, D. G., Hofman, J. M., Vaughan, J. W., & Wallach, H. (2021). Manipulating and measuring model interpretability. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–52. <https://doi.org/10.1145/3411764.3445315>

- Rathje, S., Roozenbeek, J., Van Bavel, J. J., & van der Linden, S. (2023). Accuracy and social motivations shape judgements of (mis)information. *Nature Human Behaviour*, 7, 892–903. <https://doi.org/10.1038/s41562-023-01540-w>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “why should i trust you?” explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144. <https://doi.org/10.1145/2939672.2939778>
- Schemmer, M., Kühl, N., Benz, C., Bartos, A., & Satzger, G. (2023). Appropriate reliance on AI advice: Conceptualization and the effect of explanations. *Proceedings of the 28th International Conference on Intelligent User Interfaces*, 410–422. <https://doi.org/10.1145/3581641.3584066>
- Schoeffer, J., De-Arteaga, M., & Kühl, N. (2024). Explanations, fairness, and appropriate reliance in human-AI decision-making. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–18. <https://doi.org/10.1145/3613904.3642621>
- Schoeffer, J., Jakubik, J., Vössing, M., Kühl, N., & Satzger, G. (2025). AI reliance and decision quality: Fundamentals, interdependence, and the effects of interventions. *Journal of Artificial Intelligence Research*, 82, 471–501. <https://doi.org/10.1613/jair.1.15873>
- Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 180–186. <https://doi.org/10.1145/3375627.3375830>
- Spitzer, P., Morrison, K., Turri, V., Feng, M., Perer, A., & Kühl, N. (2025). Imperfections of XAI: Phenomena influencing AI-assisted decision-making. *ACM Transactions on Interactive Intelligent Systems*. <https://doi.org/10.1145/3750052>
- Swire-Thompson, B., DeGutis, J., & Lazer, D. (2020). Searching for the backfire effect: Measurement and design considerations. *Journal of Applied Research in Memory and Cognition*, 9(3), 286–299. <https://doi.org/10.1016/j.jarmac.2020.06.006>
- Wang, X., & Yin, M. (2022). Effects of explanations in AI-assisted decision making: Principles and comparisons. *ACM Transactions on Interactive Intelligent Systems*, 12(4), 1–36. <https://doi.org/10.1145/3519266>
- Yin, M., Wortman Vaughan, J., & Wallach, H. (2019). Understanding the effect of accuracy on trust in machine learning models. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3290605.3300509>
- Zhang, Y., Liao, Q. V., & Bellamy, R. K. E. (2020). Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–17. <https://doi.org/10.1145/3351095.3372852>